

Big Data

Chapter 2

Big Data Concepts

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Practical work on NoSQL / NewSQL

Turma P1

Turma P1		
25 Fev		
Key-value	MongoDB	
	Redis	Alexandre Rafael Costa Cristóvão
	Memcached	Ana Rita Pinho Fonseca
	Etcd	Bruno Dinis Guiomar
	Hazelcast	
4 março		
Column	Cassandra	Bruno Patrício de Oliveira Fonseca
	Hbase	Francisco Miguel Nogueira Simoes
	ScyllaDB	João Pedro Silveira da Costa
	Accumulo	Juliana Silva Teixeira
	Elassandra	
11 março		
Document	Couchbase	Maria Salomé Espírito Santo Ferreira
	CouchDB	Maria Tavares Reis
	Realm	Francisco da Costa Quelhas
	RaveDB	Marta Alexandra Moreira de Oliveira
	RethinkDB	Fabrício de Lima Moreira
18 março		
Graph	Neo4J	
	ArangoDB	Rui Miguel Serralha Gonçalves
	Virtuoso	Miguel António Navarro da Costa
	OrientDB	Tiago Manuel Simões de Oliveira
	GraphDB	Daniel Costa Rodrigues
25 março		
NewSQL	CockroachDB	Tiago Ourique Cardoso
	YugaByteDB	Francisco Miguel Lopes Velhinho
	TiDB	Francisco António Ribeiro Nunes

Presentation:
20 min

Turma P2

25 Fev		
Key-value	MongoDB	
	Redis	Ana Sofia Cristóvão Ferreira da Silva
	Memcached	Francisco José Duarte Simões
	Etcd	Gonçalo Leal Gonçalves Geraldes
	Hazelcast	José Xavier Nabão
4 março		
Column	Cassandra	Alesia Lobont
	Hbase	Marina Raquel Teles Grilo Baltar
	ScyllaDB	Pedro Miguel Dinis Sequeira
	Accumulo	Rúben Soares Valente Correia
	Elassandra	
11 março		
Document	Couchbase	Rodrigo Filipe Lourenço Henriques Leitão Meireles
	CouchDB	Romeu Rafael de Melo Nogueira
	Realm	STANISLAU EBO DA SILVA
	RaveDB	Ruben João Rocha Pires Morffis Ferreiro
	RethinkDB	João Diogo Barata Soares Esteves da Fonseca
18 março		
Graph	Neo4J	Rui Damas Cruzeiro
	ArangoDB	Maksym Pata
	Virtuoso	Afonso Ferreira Sobreira
	OrientDB	BYEMBA WABULASA CHRISTIAN
	GraphDB	Pedro Antunes Pontes
25 março		
NewSQL	CockroachDB	António João Freire Fouto
	YugaByteDB	Afonso Gonçalves Rosa
	TiDB	Maria Beatriz Girão Abreu
	VoltDB	José Manuel Gouveia Moura

Presentation:
20 min

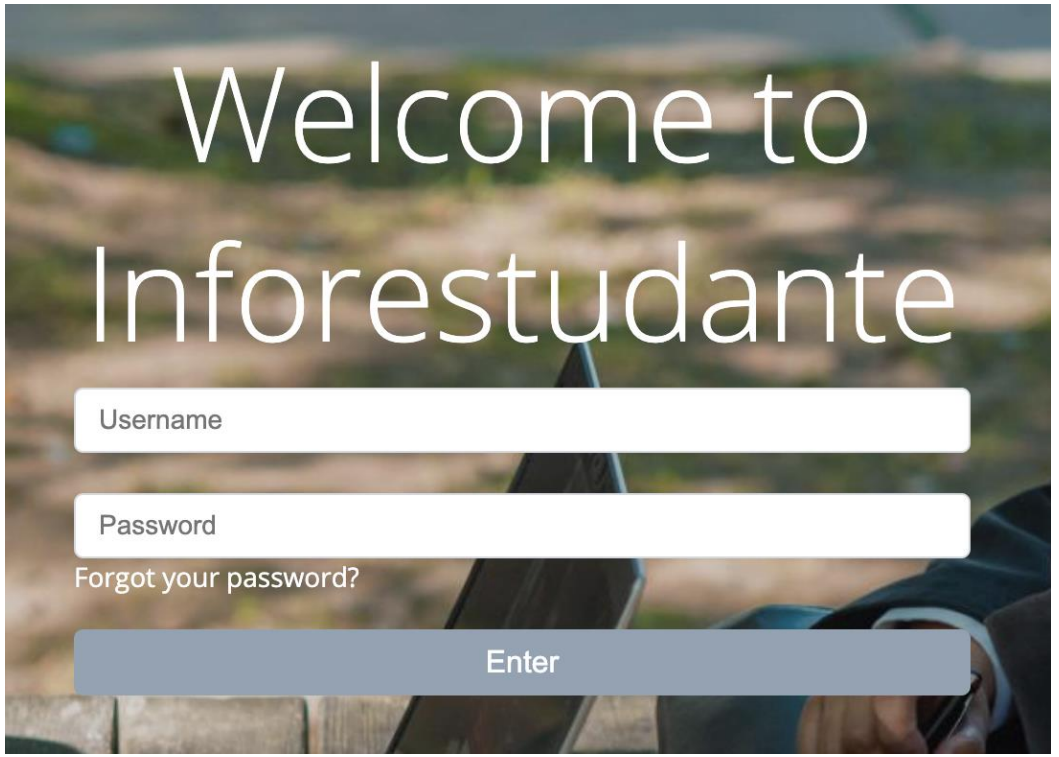
1st Practical work (Part 1) – Presentations 20min

- 25 February : Key-value
- 4 March : Column
- 11 March : Document
- 18 March : Graph
- 25 March : NewSQL

1st Practical work (**Part 2**) - Report

Report with experimental evaluation.

Deadline: **8 April at 6pm**



Practical work - Report

Title: title of the report

Abstract: Summary of the work, not exceeding 200 words.

Introduction: Introduction to the topic of NoSQL/NewSQL databases. The importance of databases and the relevance of the type of NoSQL/NewSQL database being studied should be described. This should be followed by a summary description of the essential points and contributions of this work. Finally, the structure of the report should be described.

State of the Art: Description of other work that has been done on this topic. Other NoSQL/NewSQL databases of the same type may also be mentioned.

Body of the Work (one or more sections): This section should describe all the work carried out.

Conclusions: Main conclusions of the work carried out.

References: List of references and citations in the text using APA style. References should be listed in alphabetical order.

Appendix: Description of the steps required to install the NoSQL/NewSQL database.

Practical Work on NoSQL and NewSQL Databases



In the sections of the **"Body of Work"** it is suggested that:

1) Describe the main features of the NoSQL/NewSQL DB;

2) Describe the architecture of the NoSQL/NewSQL DB;

3) Describe the practical use of the BD (which may, for example, also include the interface);

4) Experimental evaluation of the NoSQL/NewSQL DB. This experimental evaluation should include the usual operations of a database, that is, the CRUD operations. For this evaluation, the use of a public dataset is suggested. Scalability should be tested using different data sizes, for example using 1x, 10x and 100x. There are also some benchmarks for evaluating NoSQL/NewSQL DBs that can be used, such as the YCSB (Yahoo! Cloud Serving Benchmark), however they do not offer APIs for all databases;

5) Evaluation Metrics: Some of the metrics that can be evaluated are typical of any experimental evaluation: execution time, CPU time, RAM memory used, disk space occupied, etc.

6) Comparison with a usual relational DB: MySQL, PostgreSQL, MariaDB or alternatively comparison with a DB of the same type.

Template: https://www.mdpi.com/journal/BDCC/instructions



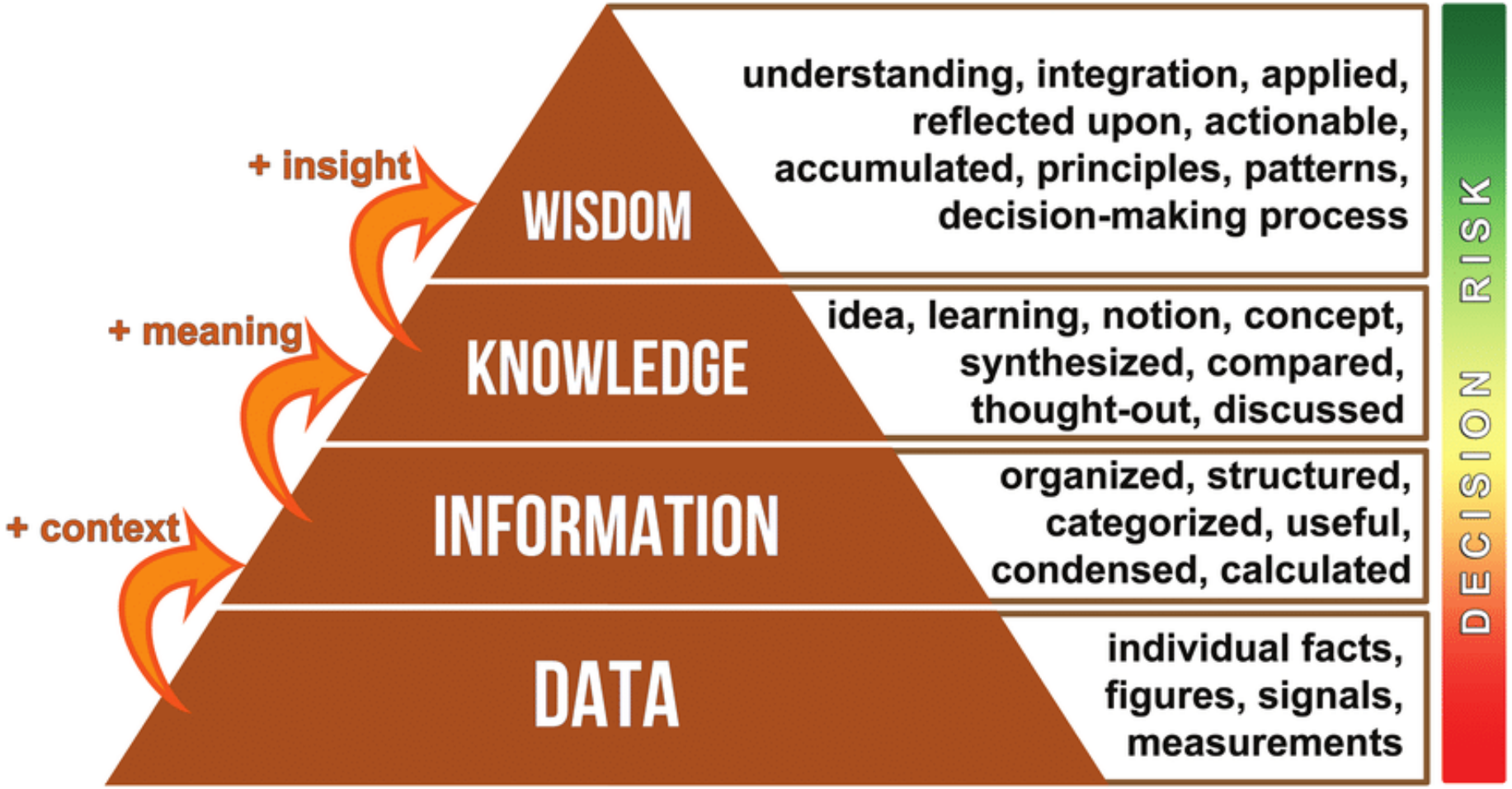
Deadline: the report must be placed on **NONIO by 6 pm on April 8th**.

Agenda

- What is Analytics?
- What is Big Data?
- Evolution of Big Data and Data Analytics
- Characteristics of Big Data
- Domain Specific Examples of Big Data
- Failure of Traditional Database in Handling Big Data
- Eras of Big Data
- Analytics Flow for Big Data
- Big Data Stack

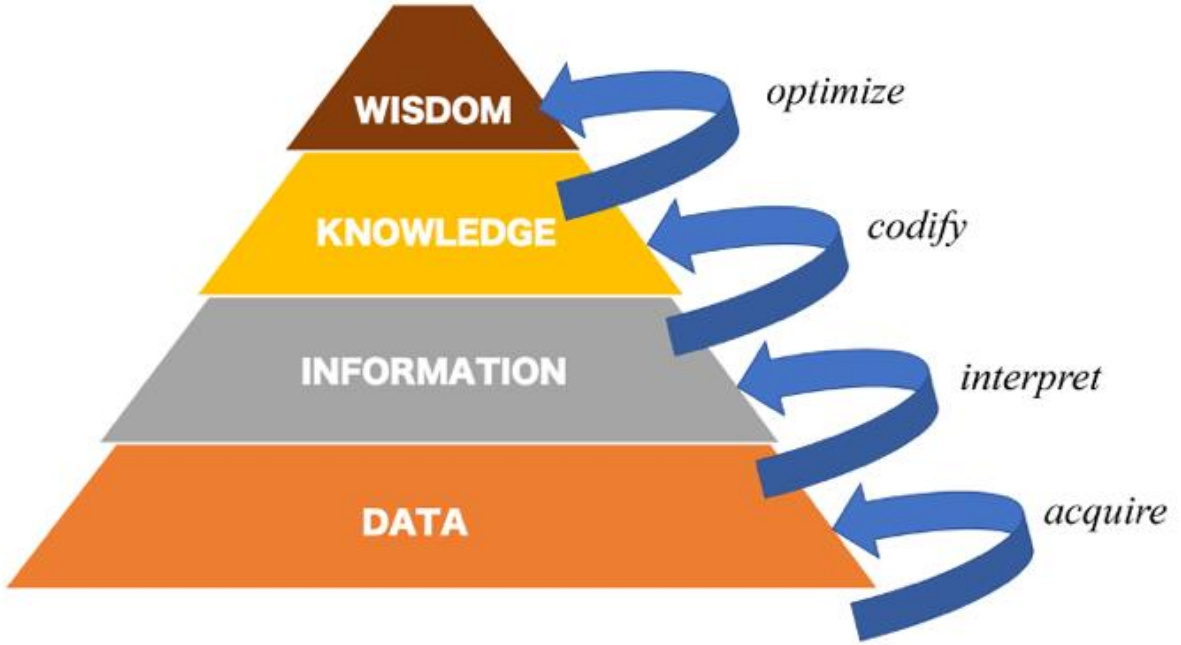
DIKW ?

Data, Information, Knowledge and Wisdom

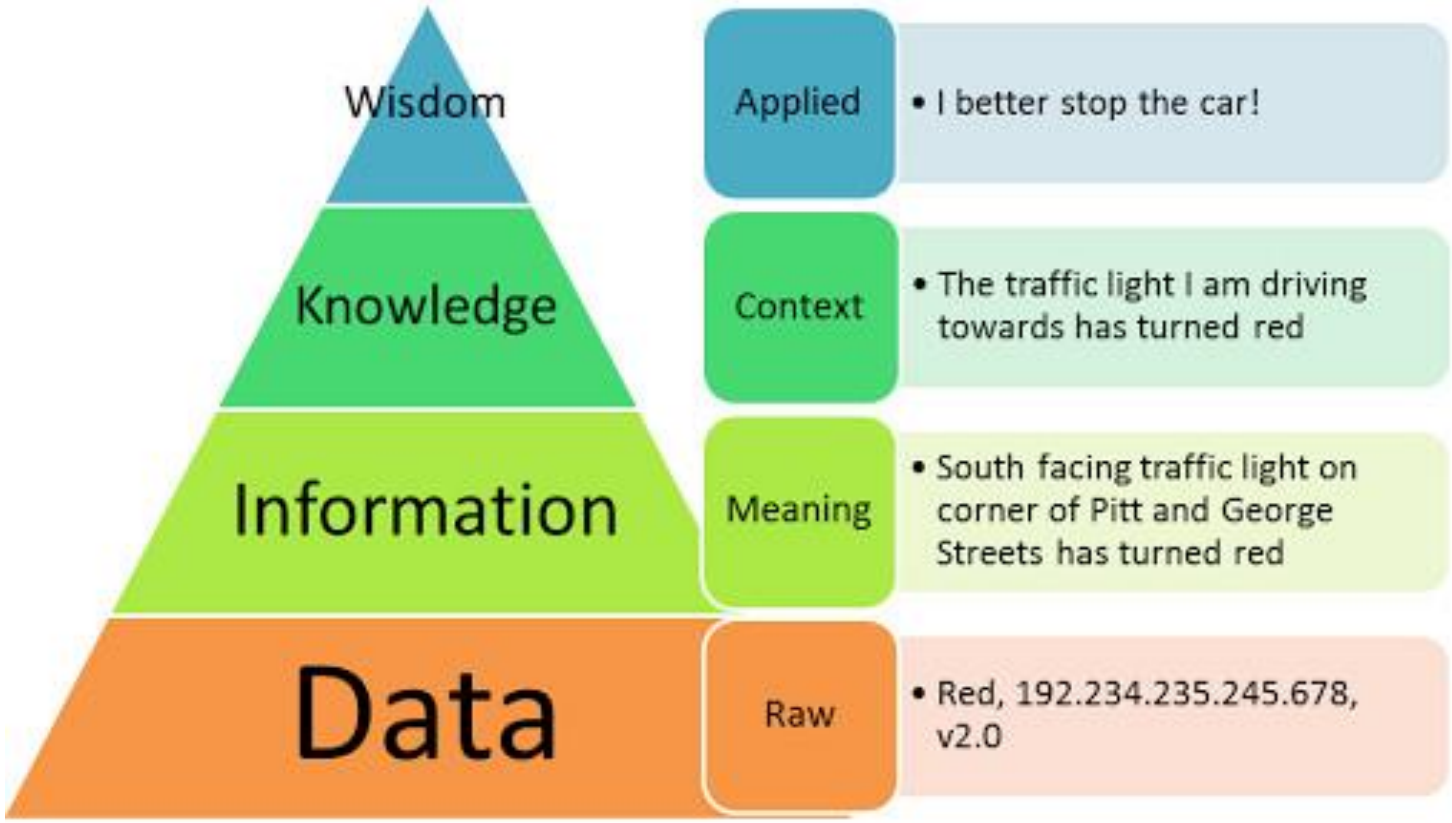


Data, Information, Knowledge and Wisdom

- Data … You acquire it through means at your disposal.
- Information … You interpret data into information.
- Knowledge … You codify information into knowledge.
- Wisdom … You optimize knowledge into wisdom.



Data, Information, Knowledge and Wisdom



Product Example: Mobile App Development

Mobile apps collect user data, and can get additional feedback data from users, but the product manager's end goal is to make decisions to improve the app.

- **Data**: The raw data consists of individual user interactions with the app, such as button clicks, screen views, and time spent in the app, as well as user-submitted feedback through reviews, surveys, and support tickets.
- **Information**: App usage and user feedback data is organized and structured, providing metrics like average session duration, feature usage frequency, user retention rates.
- **Knowledge**: Analyzing and interpreting information from app usage and user feedback uncovers patterns such as frequently requested features, causes of user frustration, and key factors driving user engagement and loyalty.
- **Wisdom**: Understanding user needs, preferences, and pain points lets the product manager make informed decisions to prioritize feature development, enhance user experience, and allocate resources effectively to maximize user satisfaction and retention.

Everyday Life Example: Fitness tracking

Fitness tracking devices collect your health and activity data, but your end goal is to use that to make decisions about how to train or how to manage your health.

- **Wisdom**: Understanding these patterns lets you make informed decisions about adjusting your exercise routine, sleep habits, and other lifestyle factors to improve your health and fitness.

⇧

- **Knowledge**: Analyzing and interpreting the information may reveal patterns, such as increased step count leading to improved sleep quality or a correlation between heart rate and workout intensity.

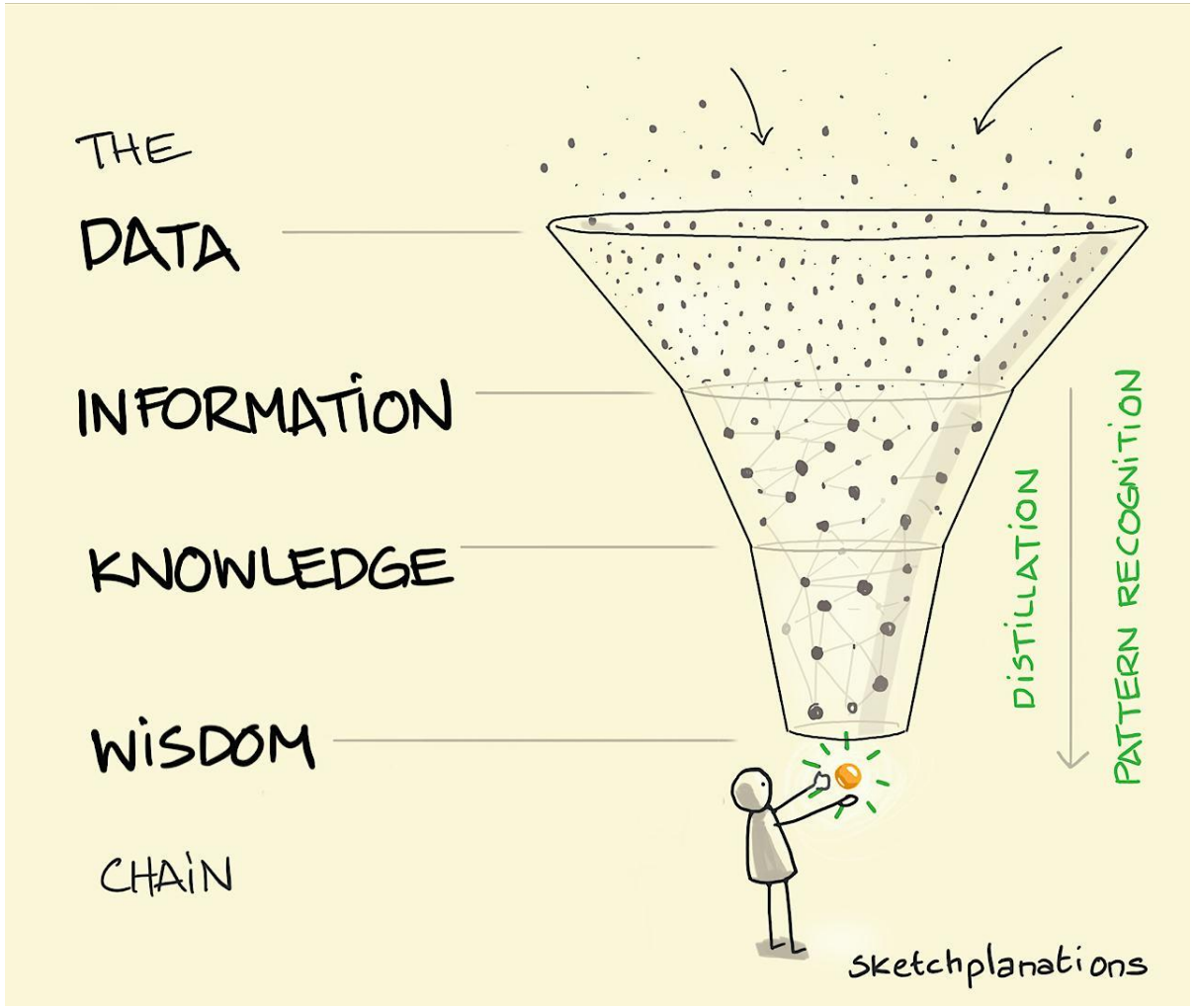
⇧

- **Information**: The smartwatch app organizes and structures the data, displaying it in a comprehensible format, such as daily step count, average heart rate, and hours of sleep per night.

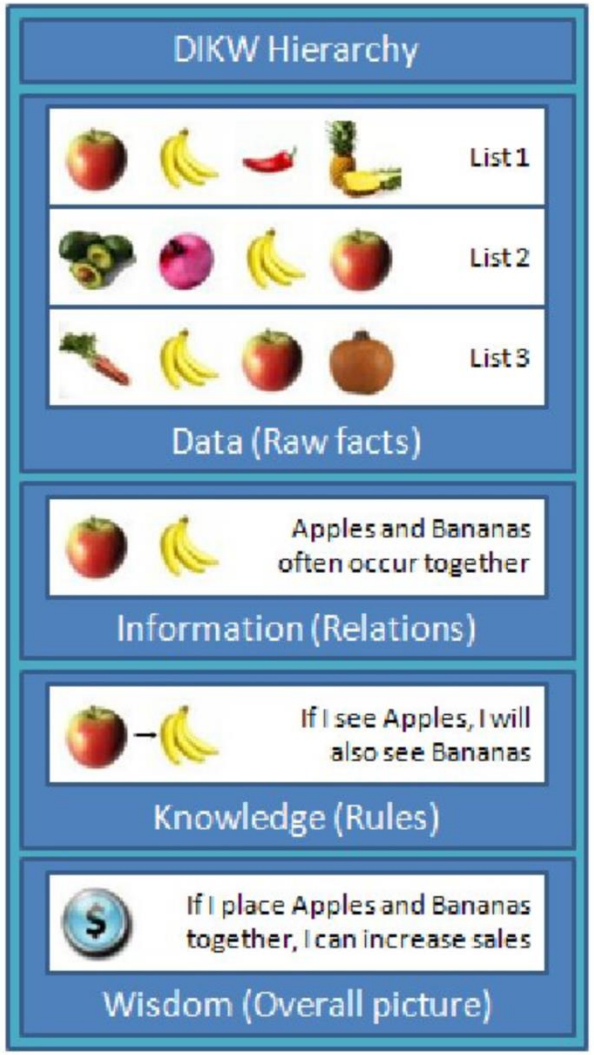
⇧

- **Data**: The smartwatch collects raw data such as the number of steps taken, heart rate, and sleep duration

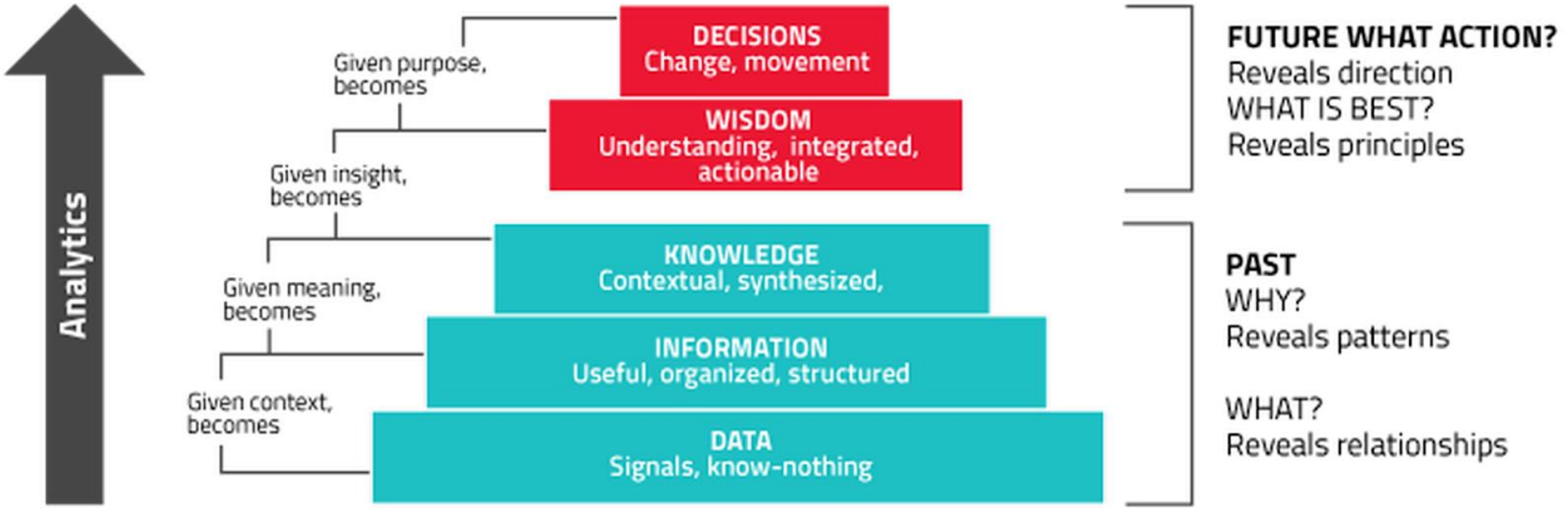
Data, Information, Knowledge and Wisdom



Data, Information, Knowledge and Wisdom



DIKW pyramid in decision-making



What is Analytics?

- Analytics is a broad term that encompasses the processes, technologies, frameworks and algorithms to extract meaningful insights from data.
- Raw data in itself does not have a meaning until it is contextualized and processed into useful information.
- Analytics is this process of extracting and creating information from raw data by filtering, processing, categorizing, condensing and contextualizing the data.
- This information obtained is then organized and structured to infer knowledge about the system and/or its users, its environment, and its operations and progress towards its objectives, thus making the systems smarter and more efficient.

What is Analytics?

The goals of the analytics task may be:

1) **to predict something**

   (for example, whether a transaction is a fraud or not, whether it will rain on a particular day, or whether a tumour is benign or malignant)

2) **to find patterns in the data**

   (for example, finding the top 10 coldest days in the year, finding which pages are visited the most on a particular website, or finding the most searched celebrity in a particular year)

3) **finding relationships in the data**

   (for example, finding similar news articles, finding similar patients in an electronic health record system, finding related products on an eCommerce website, finding similar images, or finding correlation between news items and stock prices).

Types of Analytics

- Descriptive Analytics
- Diagnostic Analytics
- Predictive Analytics
- Prescriptive Analytics

Types of Analytics

Descriptive Analytics

- Descriptive analytics involves analyzing historical data to present it in a summarized form that can be easily interpreted. Descriptive analytics aims to answer - **What has happened?**
- Much of the analysis done today is descriptive analytics using statistical functions such as Count, Max, Min, Average, Percentage.

Diagnostic Analytics

- Diagnostic analytics comprises analysis of past data to diagnose the reasons as to why certain events happened. Diagnostic analytics aims to answer - **Why did it happen?**
- Consider an example of a system that collects and analyzes sensor data from machines for monitoring their health and predicting failures. While descriptive analytics can be useful for summarizing the data by computing various statistics (such as mean, minimum, maximum, variance, or top-N), diagnostic analytics can provide more insights into why certain a fault has occurred based on the patterns in the sensor data for previous faults.

Types of Analytics

Predictive Analytics

- Predictive analytics comprises predicting the occurrence of an event or the likely outcome of an event or forecasting the future values using prediction models. Predictive analytics aims to answer - **What is likely to happen?**
- For example, predictive analytics can be used for predicting when a fault will occur in a machine, predicting whether a tumour is benign or malignant, predicting the occurrence of natural emergency (events such as forest fires or river floods) or forecasting the pollution levels.

Prescriptive Analytics

- While predictive analytics uses prediction models to predict the likely outcome of an event, prescriptive analytics uses multiple prediction models to predict various outcomes and the best course of action for each outcome. Prescriptive analytics aims to answer - **What can we do to make it happen?**
- Prescriptive Analytics can predict the possible outcomes based on the current choice of actions. We can consider prescriptive analytics as a type of analytics that uses different prediction models for different inputs. Prescriptive analytics prescribes actions or the best option to follow from the available options.
- For example, prescriptive analytics can be used to prescribe the best medicine for treatment of a patient based on the outcomes of various medicines for similar patients.

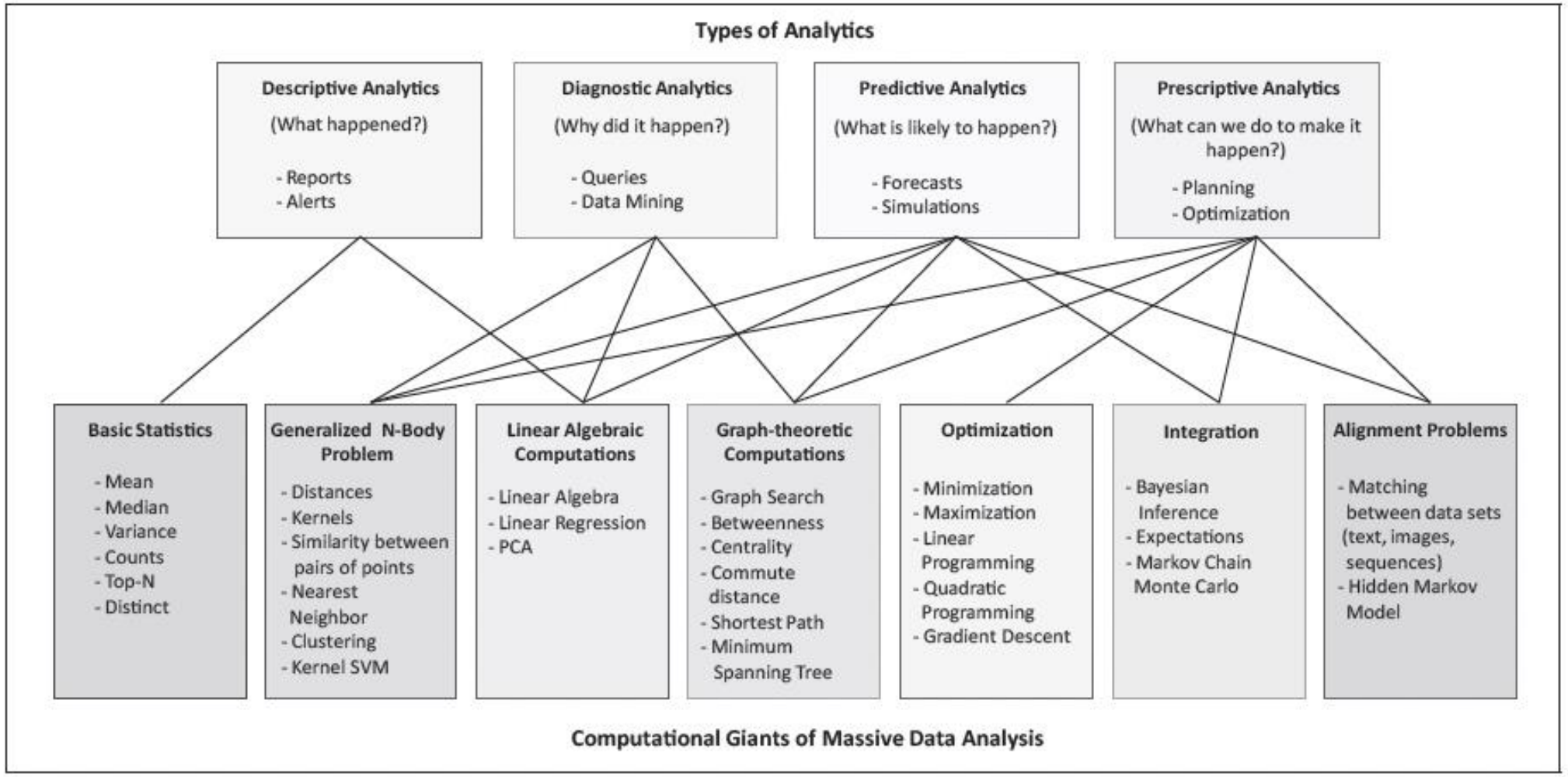
What is Analytics?

- The National Research Council [1] has done a characterization of computational tasks for massive data analysis (called the seven "giants"). These computational tasks include:
  - **(1) Basics Statistics**
  - **(2) Generalized N-Body Problems**
  - **(3) Linear Algebraic Computations**
  - **(4) Graph-Theoretic Computations**
  - **(5) Optimization**
  - **(6) Integration**
  - **(7) Alignment Problems.**

  - **Computational Giants of Massive Data Analysis**
- This characterization of computational tasks aims to provide a taxonomy of tasks that have proved to be useful in data analysis and grouping them roughly according to mathematical structure and computational strategy.

[1] National Research Council, Frontiers in Massive Data Analysis, The National Academies Press, 2013.

Types of Analytics



Evolution of Big Data and Data Analytics

- While the emergence of big data occurred only recently, the act of gathering and storing large amounts of data dates back to the early 1950s when the first commercial mainframe computers were introduced.
- During the period between the early 1950s to mid-1990s, data grew relatively slowly due to the high cost of computers, storage, and data networks.
- Data during this period were highly structured, mainly to support operational and transactional information systems.
- The advent of the world wide web (WWW) in the early 1990s led to the explosive growth of data and the development of big data analytics. Since the advent of the WWW, big data and data analytics have evolved through three major stages.

Big Data 1.0 (1994 - 2004)

- Big Data 1.0 coincides with the advent of e-commerce in 1994, during which time online firms were the main contributors of the web content.
- User-generated content was only a marginal part of web content due to the technical limitation of web applications.
- In this era, web mining techniques were developed to analyze users' online activities.
- Web mining can be divided into three different types:
  - web usage mining
  - web structure mining
  - web content mining.

Big Data 2.0 (2005 - 2014)

- Big Data 2.0 is driven by Web 2.0 and the social media phenomenon.
- Web 2.0 refers to a web paradigm that evolved from the web technologies of the 1990s and allowed web users to interact with web-sites and contribute their own content to the web-sites.
- Social media embodied the principles of Web 2.0 and created a paradigm shift in the way organizations operate and collaborate.
- Social media analytics support social media content mining, usage mining, and structure mining activities
  - Social media analytics focus on two types of analysis:
    - sentiment analysis
    - social network analysis.

Big Data 3.0 (2015 -)

- Big Data 3.0 encompasses data from Big Data 1.0 and Big Data 2.0.
- The main contributors of Big Data 3.0 are the IoT applications that generate data in the form of images, audio, and video.
- The IoT refers to a technology environment in which devices and sensors have unique identifiers with the ability to share data and collaborate over the Internet even without any human intervention.
- For many IoT applications, the analysis increasingly is performed by sensors at the source of data gathering. This trend is leading to a new field known as streaming analytics. Streaming analytics continuously extract information from the streaming data.
- Unlike social media analytics used in a batch mode for the analysis of stored data, streaming analytics involve real-time event analysis to discover patterns of interest as data is being collected or generated.
- Streaming analytics are used not just for monitoring existing conditions but also for predicting future events.

What is Big Data?

- Big data is defined as collections of datasets whose volume, velocity or variety is so large that it is difficult to store, manage, process and analyze the data using traditional databases and data processing tools.

- In the recent years, there has been an exponential growth in the both structured and unstructured data generated by information technology, industrial, healthcare, Internet of Things, and other systems.

What is Big Data? (2025)

- Approximately 9–10 petabytes (10^15 bytes) of data are generated every minute by 2025 forecasts, rising from around 6.4 petabytes/min in 2021.

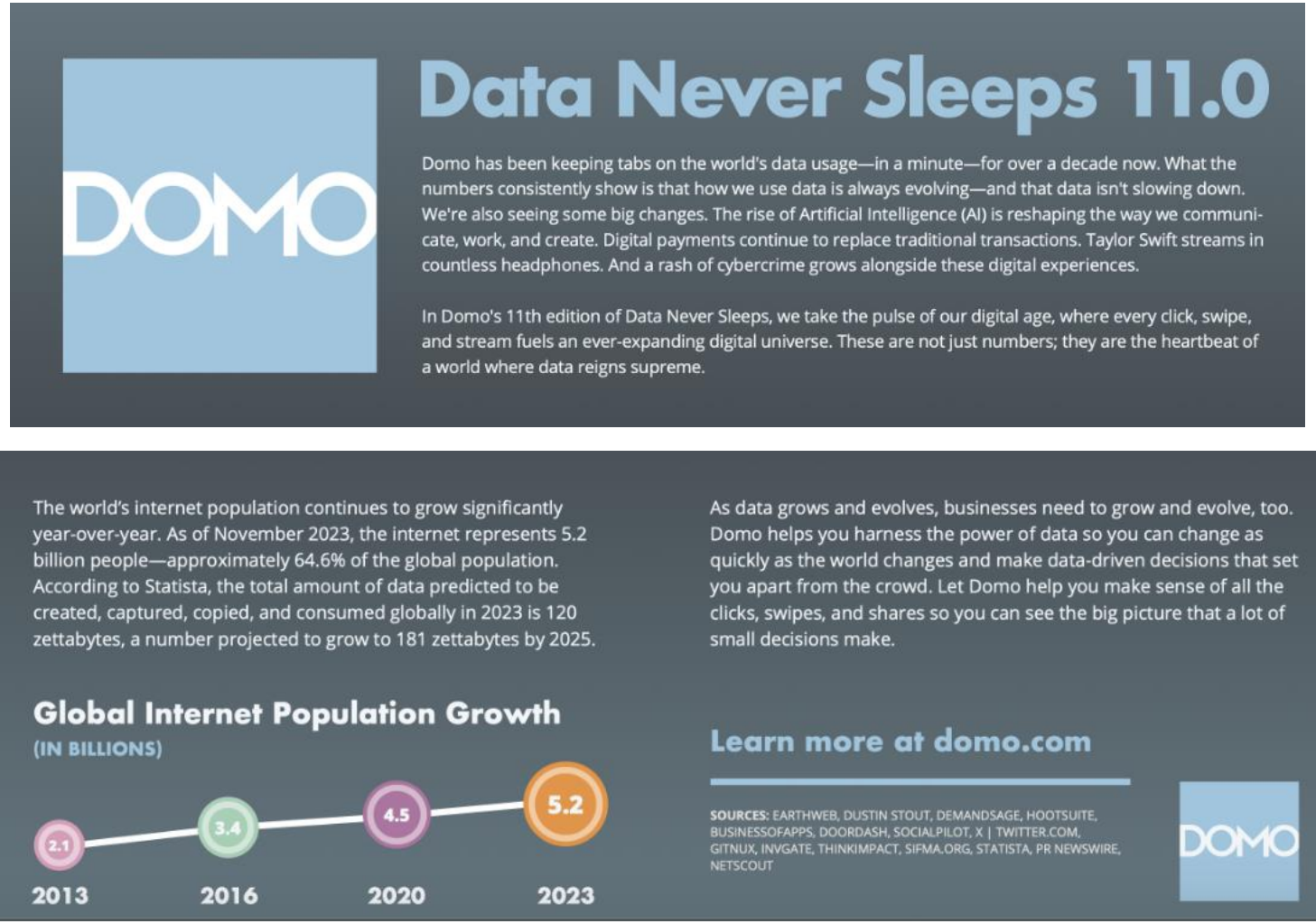
- This explosion of data:

  2.5 quintillion bytes per day (2.5x10^18 or 2.5 exabytes)

  is driven by millions of global users watching 4.1 million YouTube videos, sending 231 million emails, and performing over 5.7 million Google searches every 60 seconds.

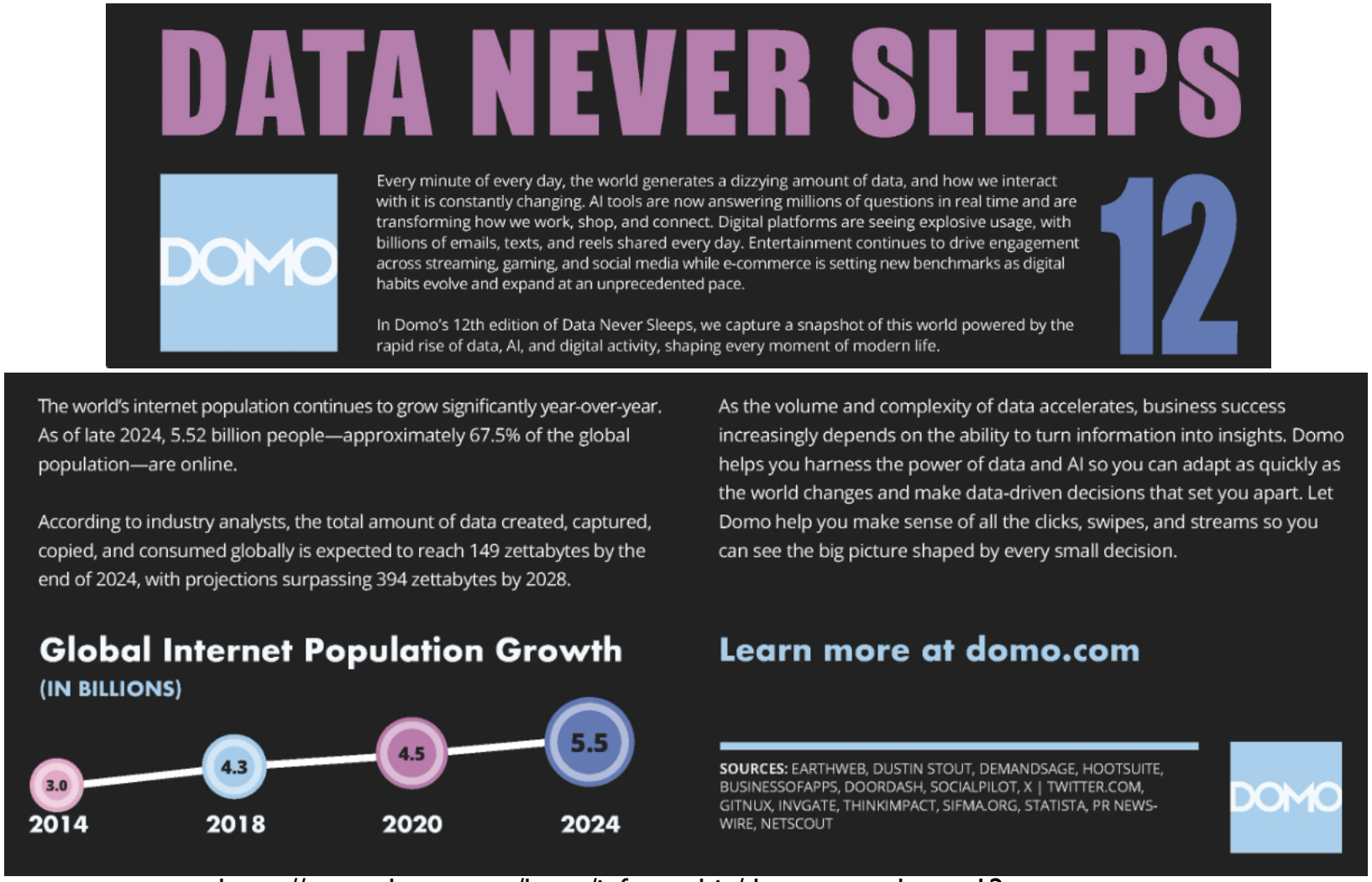
Source: Google IA

What is Big Data? (2023)



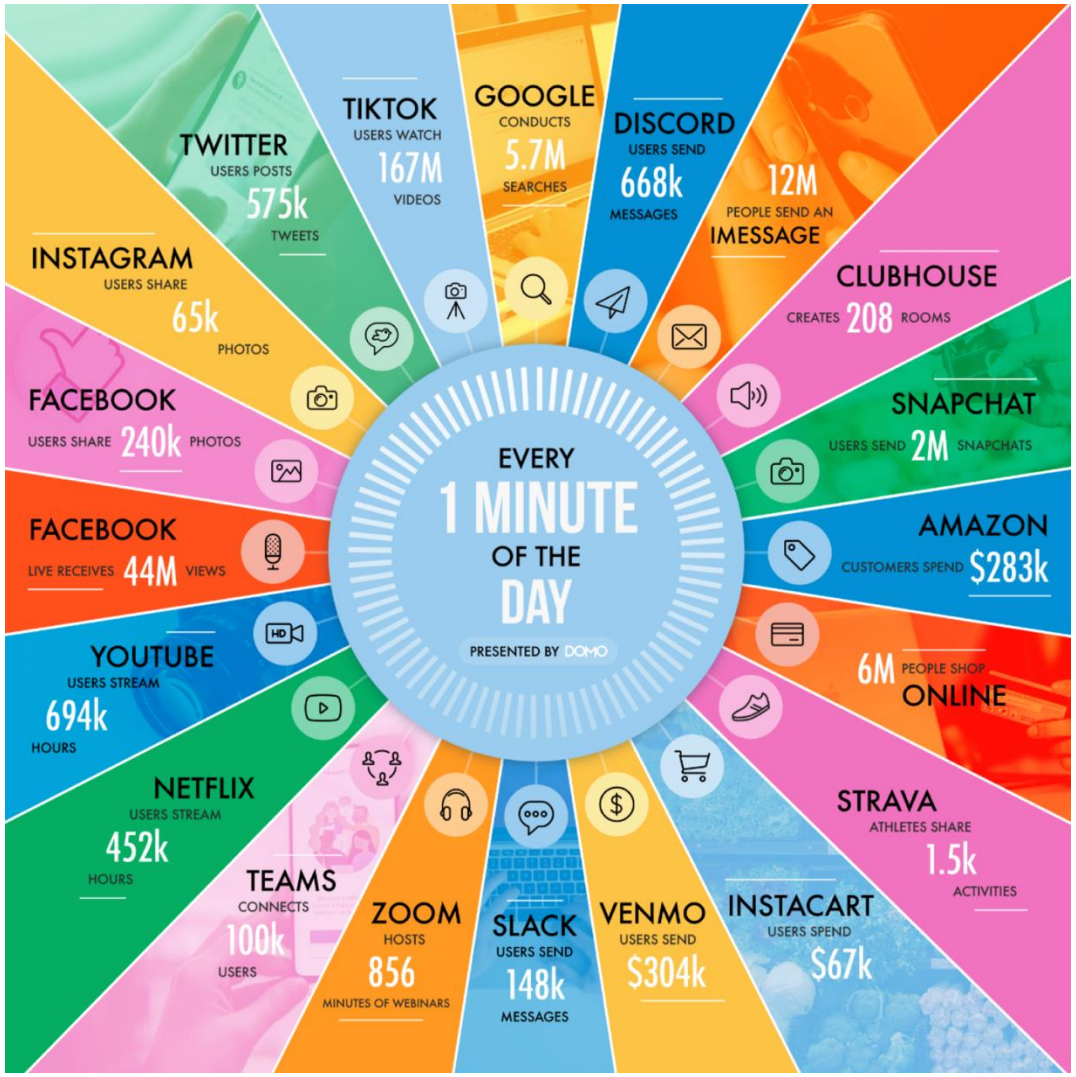
https://www.domo.com/learn/infographic/data-never-sleeps-11

What is Big Data? (2024)

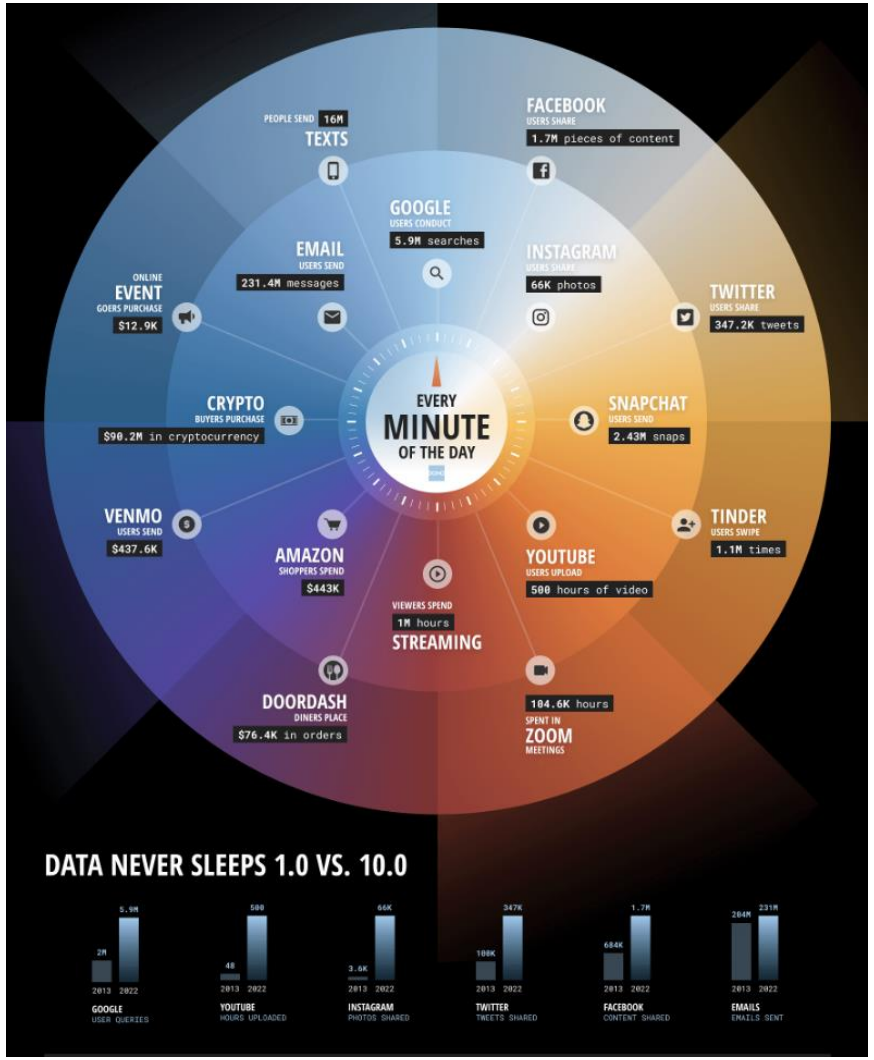


https://www.domo.com/learn/infographic/data-never-sleeps-12

What is Big Data? (2021)



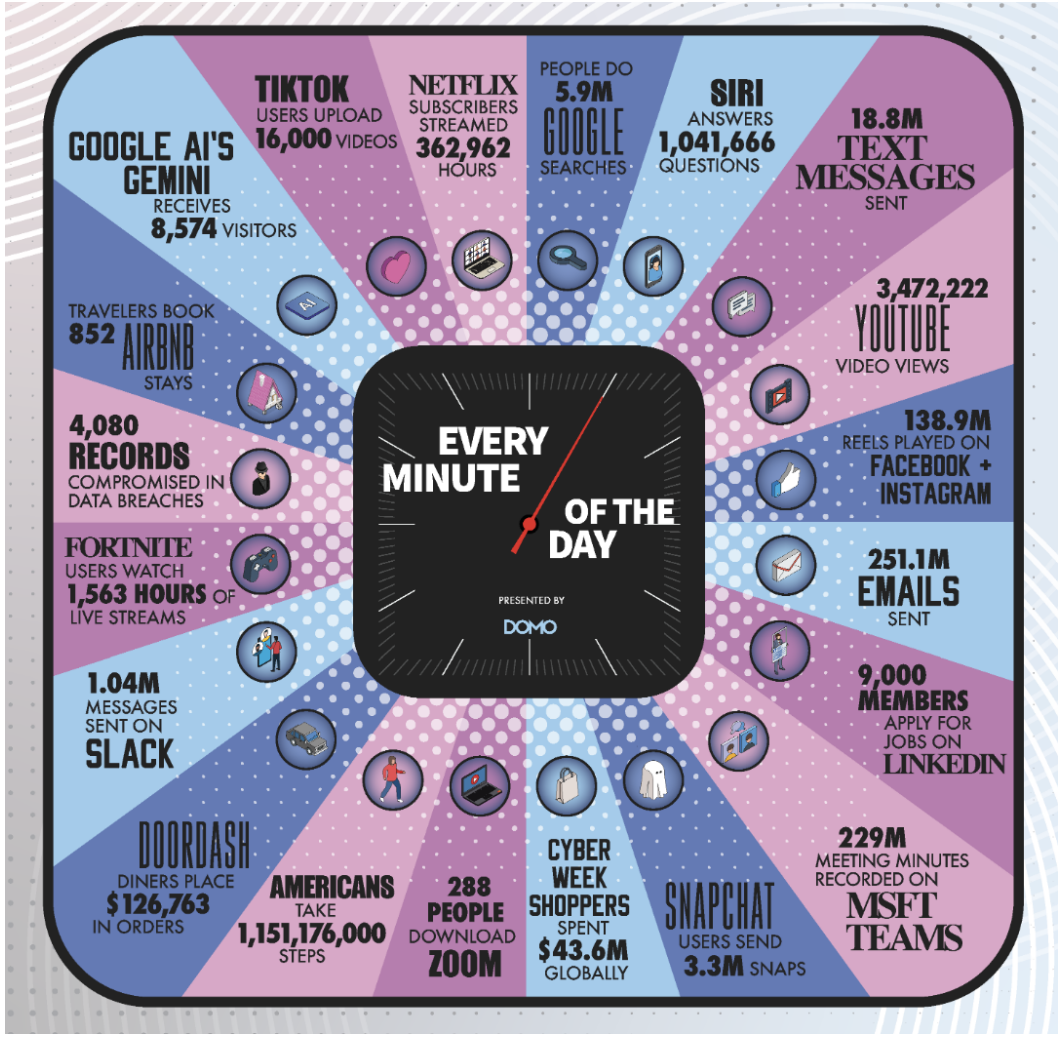
What is Big Data? (2022 version)



What is Big Data? (2023)

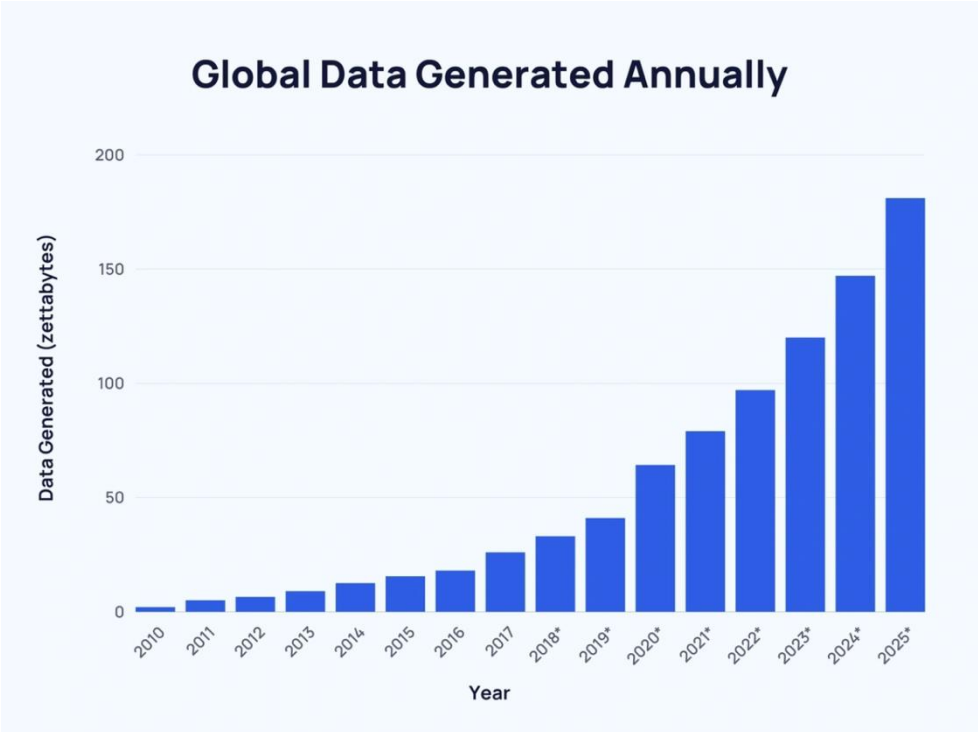


What is Big Data? (2024)



Data Creation Growth Projections

- In the space of 13 years, this figure has increased by an estimated **74x** from just **2 zettabytes** (10^{21}) in 2010.
- The **120 zettabytes** generated in 2023 are expected to increase by **over 150%** in 2025, hitting **181 zettabytes**.



Source: https://explodingtopics.com/blog/data-generated-per-day

What is Big Data?

- Big Data has the potential to power next generation of smart applications that will leverage the power of the data to make the applications intelligent.

- Applications of big data cover a wide range of domains such as web, retail and marketing, banking and financial, industrial, healthcare, environmental, Internet of Things and cyber-physical systems.

What is Big Data?

- Big Data analytics deals with collection, storage, processing and analysis of this massive scale data. Specialized tools and frameworks are required for big data analysis when:

  1) the volume of data involved is so large that it is difficult to store, process and analyze data on a single machine,

  2) the velocity of data is very high and the data needs to be analyzed in real-time,

  3) there is variety of data involved, which can be structured, unstructured or semi-structured, and is collected from multiple data sources,

What is Big Data?

- Specialized tools and frameworks are required for big data analysis when (cont.):

  1) 2) 3)

  4) various types of analytics need to be performed to extract value from the data such as descriptive, diagnostic, predictive and prescriptive analytics.

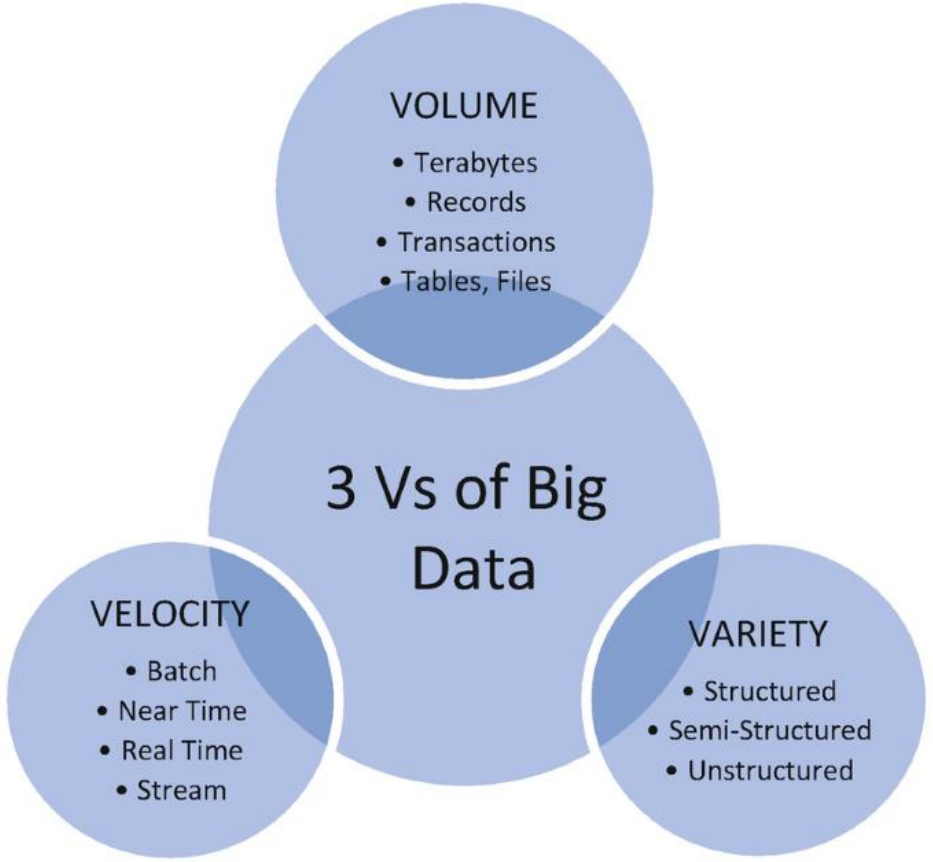
- Big Data tools and frameworks have distributed and parallel processing architectures and can leverage the storage and computational resources of a large cluster of machines.

What is Big Data?

Some examples of Big Data :

- Data generated by social networks including text, images, audio, and video data
- Click-stream data generated by web applications such as e-Commerce to analyze user behavior
- Machine sensor data collected from sensors embedded in industrial and energy systems for monitoring their health and detecting failures
- Healthcare data collected in electronic health record (EHR) systems
- Logs generated by web applications
- Stock markets data
- Transactional data generated by banking and financial applications.

Characteristics of Big Data: 3 V's



1024 GB = 1 terabyte; 1024 TB = 1 petabyte; 1024 PB = 1 exabyte;
1024 EB = 1 zettabyte; 1024 ZB = 1 yottabyte

Characteristics of Big Data

Volume

- Big data is a form of data whose volume is so large that it would not fit on a single machine, therefore specialized tools and frameworks are required to store process and analyze such data.
- For example, social media applications process billions of messages everyday, industrial and energy systems can generate terabytes of sensor data everyday, cab aggregation applications can process millions of transactions in a day, etc.
- The volumes of data generated by modern IT, industrial, healthcare, Internet of Things, and other systems is growing exponentially driven by the lowering costs of data storage and processing architectures and the need to extract valuable insights from the data to improve business processes, efficiency and service to consumers.
- Though there is no fixed threshold for the volume of data to be considered as big data, however, typically, the term big data is used for massive scale data that is difficult to store, manage and process using traditional databases and data processing architectures.

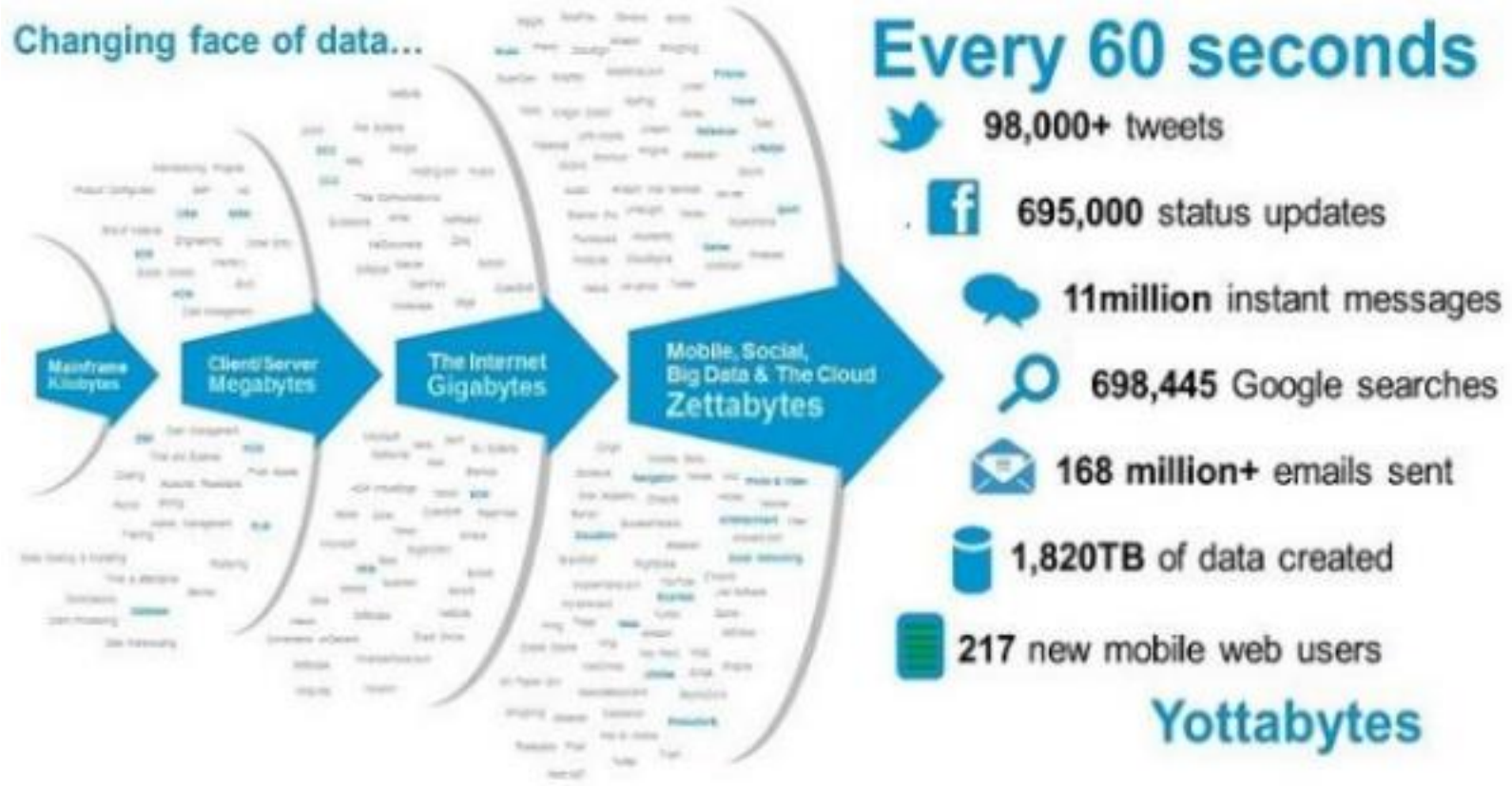
Characteristics of Big Data

Velocity

- Velocity of data refers to how fast the data is generated. Data generated by certain sources can arrive at very high velocities, for example, social media data or sensor data.
- Velocity is another important characteristic of big data and the primary reason for the exponential growth of data.
- High velocity of data results in the volume of data accumulated to become very large, in short span of time. Some applications can have strict deadlines for data analysis (such as trading or online fraud detection) and the data needs to be analyzed in real-time.
- Specialized tools are required to ingest such high velocity data into the big data infrastructure and analyze the data in real-time.

Characteristics of Big Data

Velocity



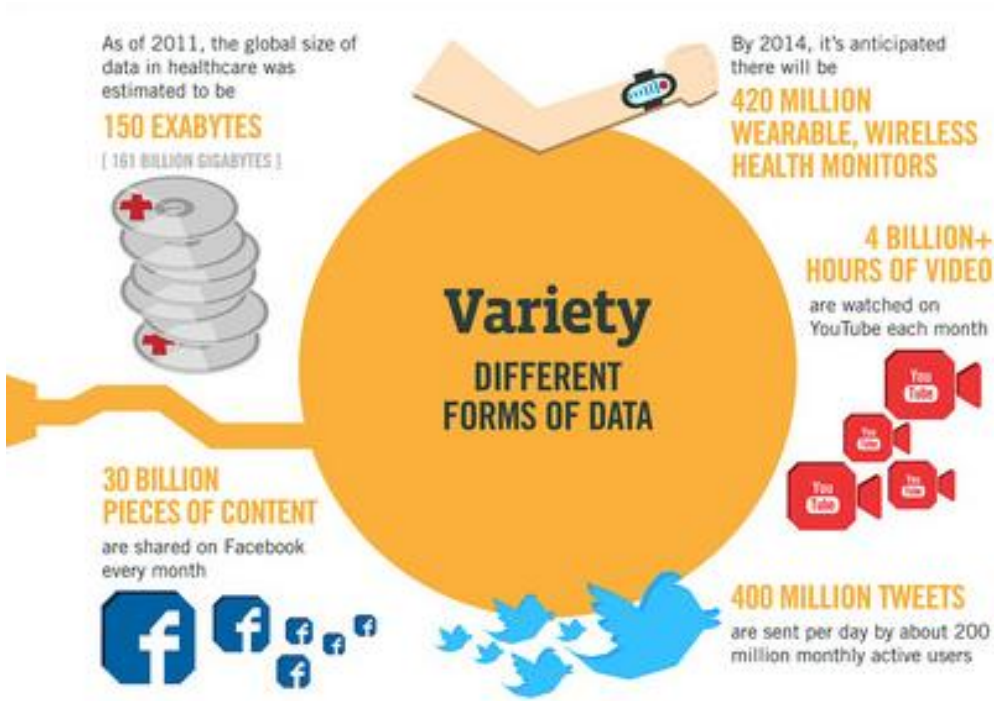
Characteristics of Big Data

Variety

- Variety refers to the forms of the data.
- Big data comes in different forms such as structured, unstructured or semi-structured, including text data, image, audio, video and sensor data.
- Big data systems need to be flexible enough to handle such variety of data.

Characteristics of Big Data

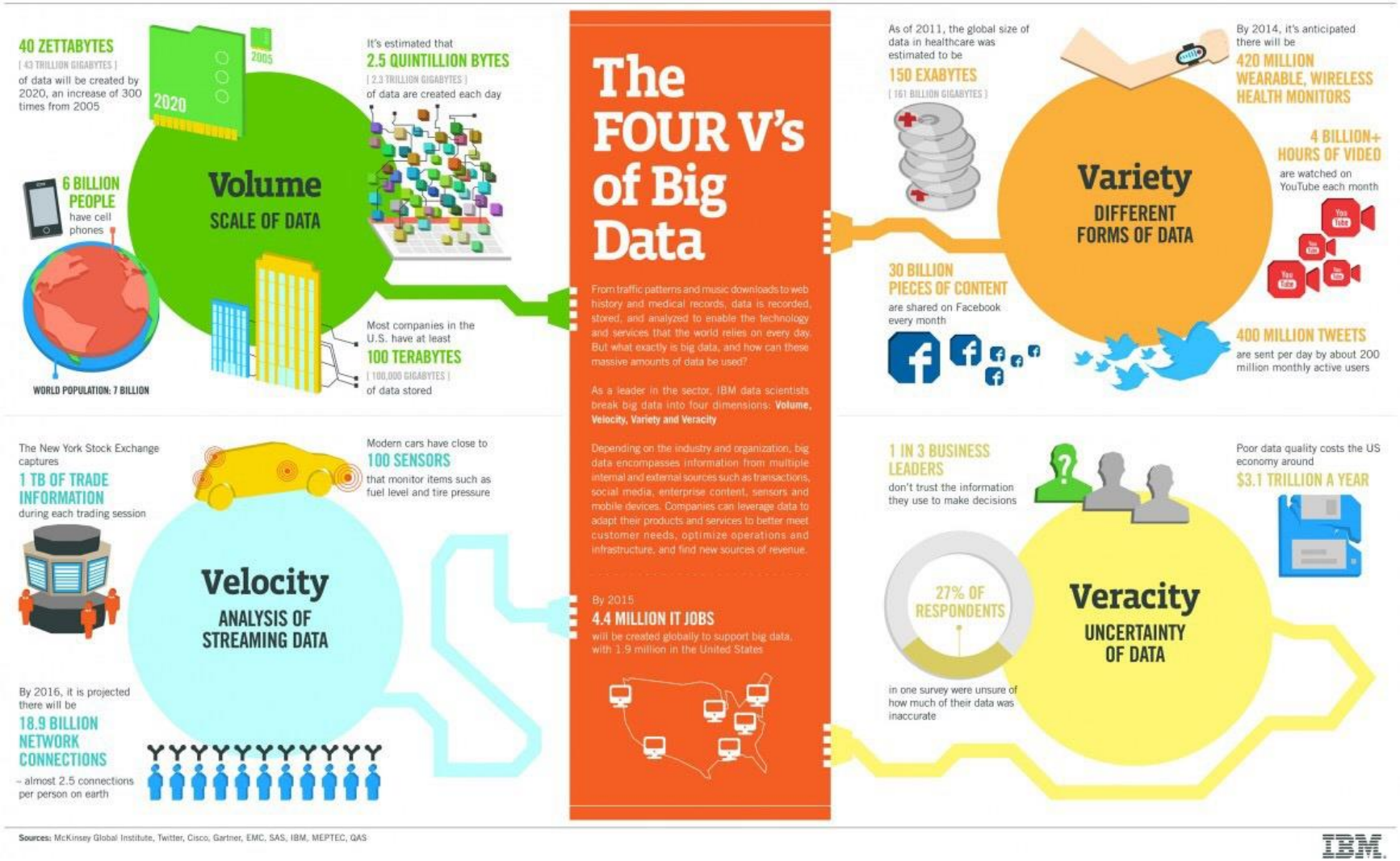
Variety



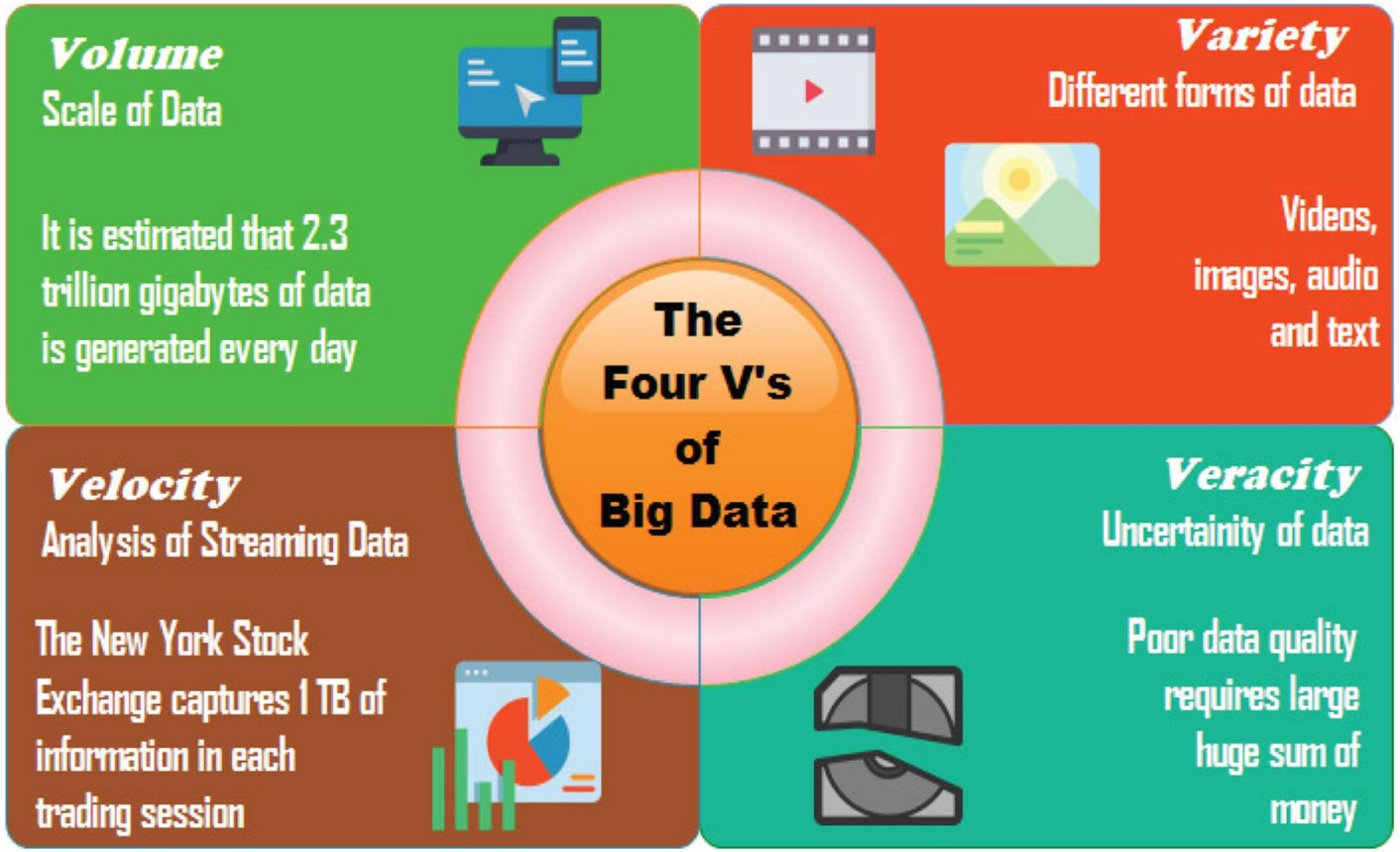


Structured Data | Unstructured Data | Semi-Structured Data

4 V's of Big Data



4 V's of Big Data



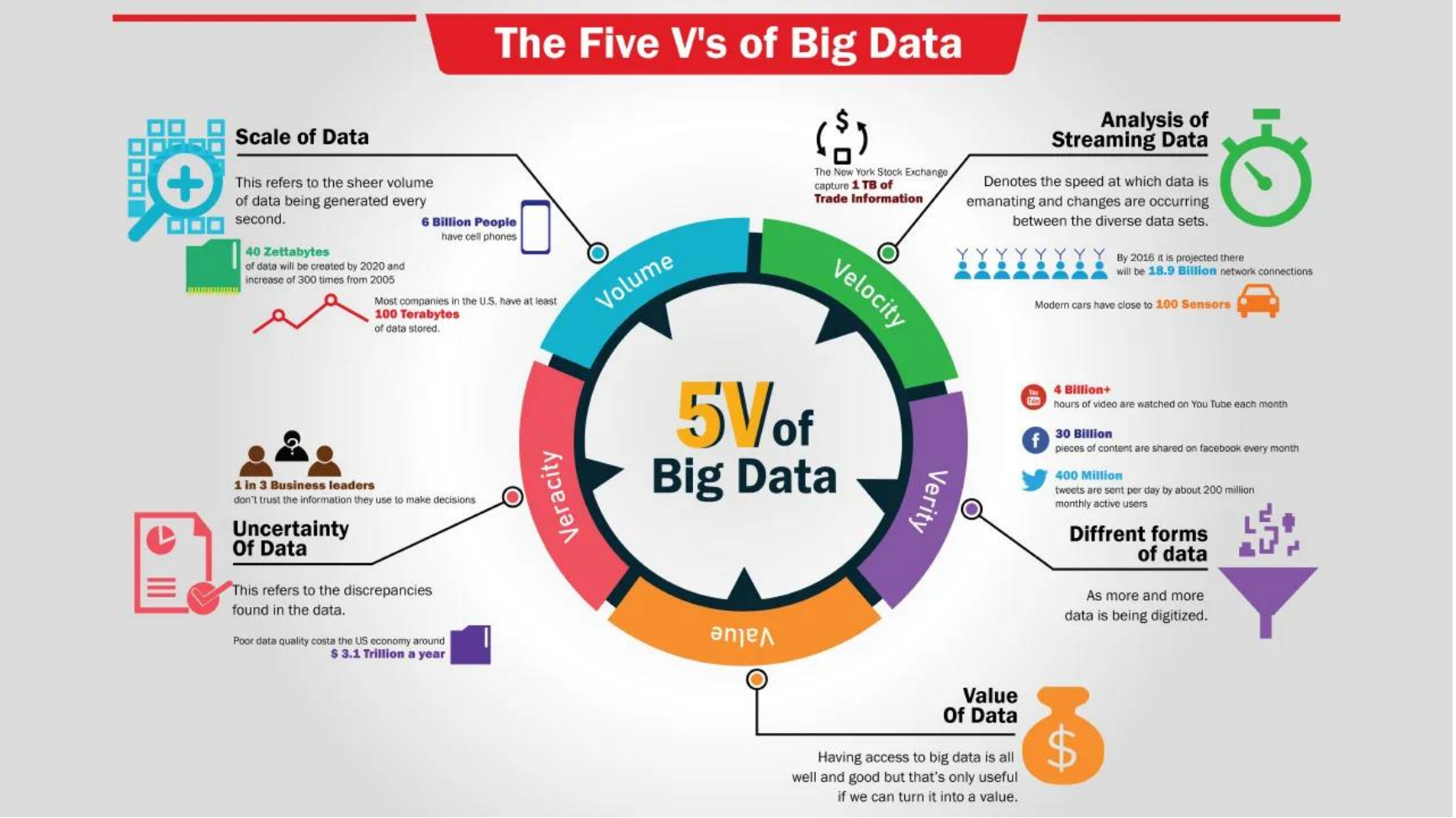
Softw Pract Exp, Volume: 52, Issue: 3, Pages: 658-688, First published: 12 August 2020, DOI: (10.1002/spe.2876)

Characteristics of Big Data

Veracity

- Veracity refers to how accurate is the data.
- To extract value from the data, the data needs to be cleaned to remove noise.
- Data-driven applications can only benefit from big data if the data is meaningful and accurate.
- Therefore, cleansing of data is important so that incorrect and faulty data can be filtered out.
  - Examples

5 V's of Big Data

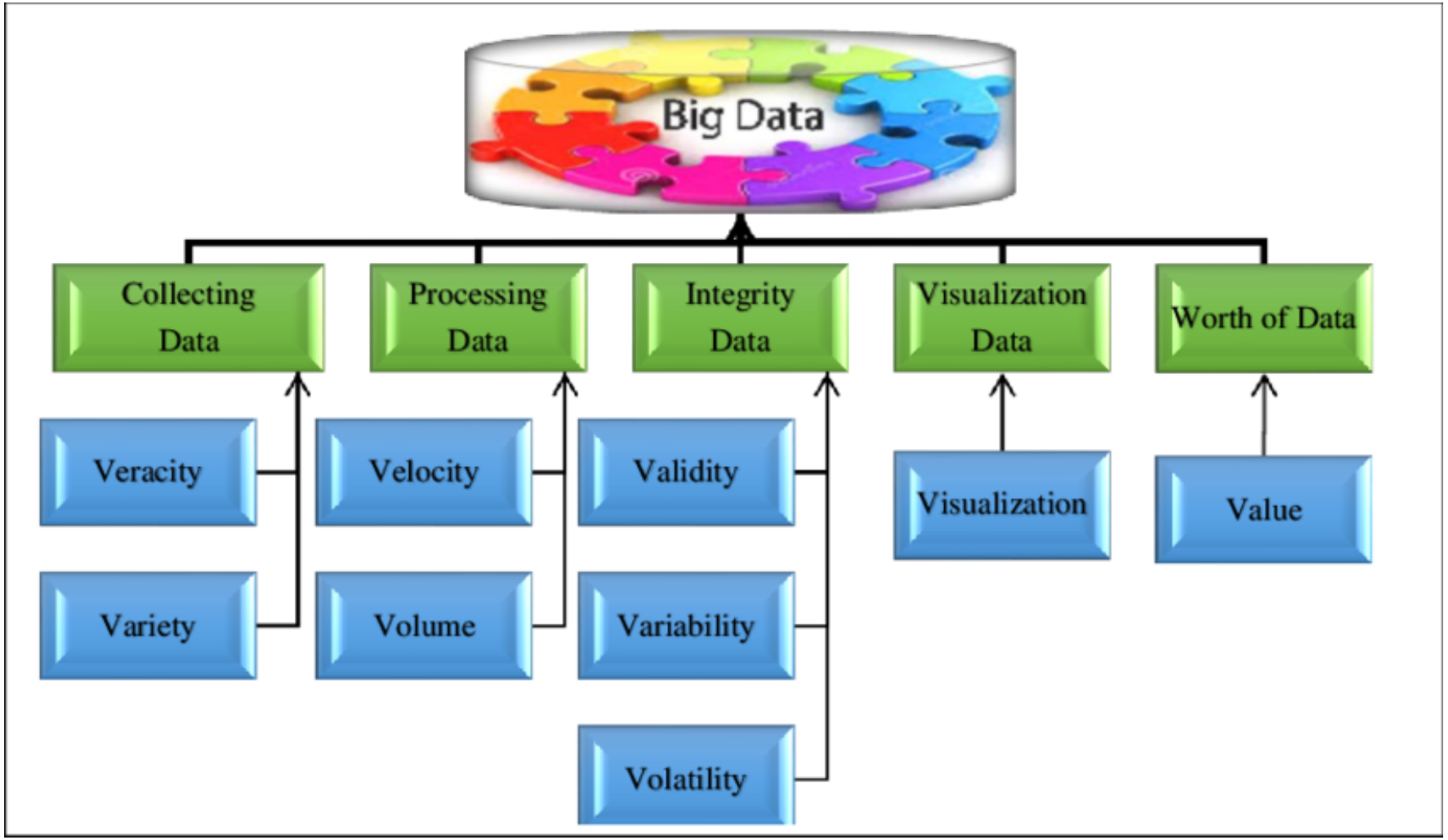


Characteristics of Big Data

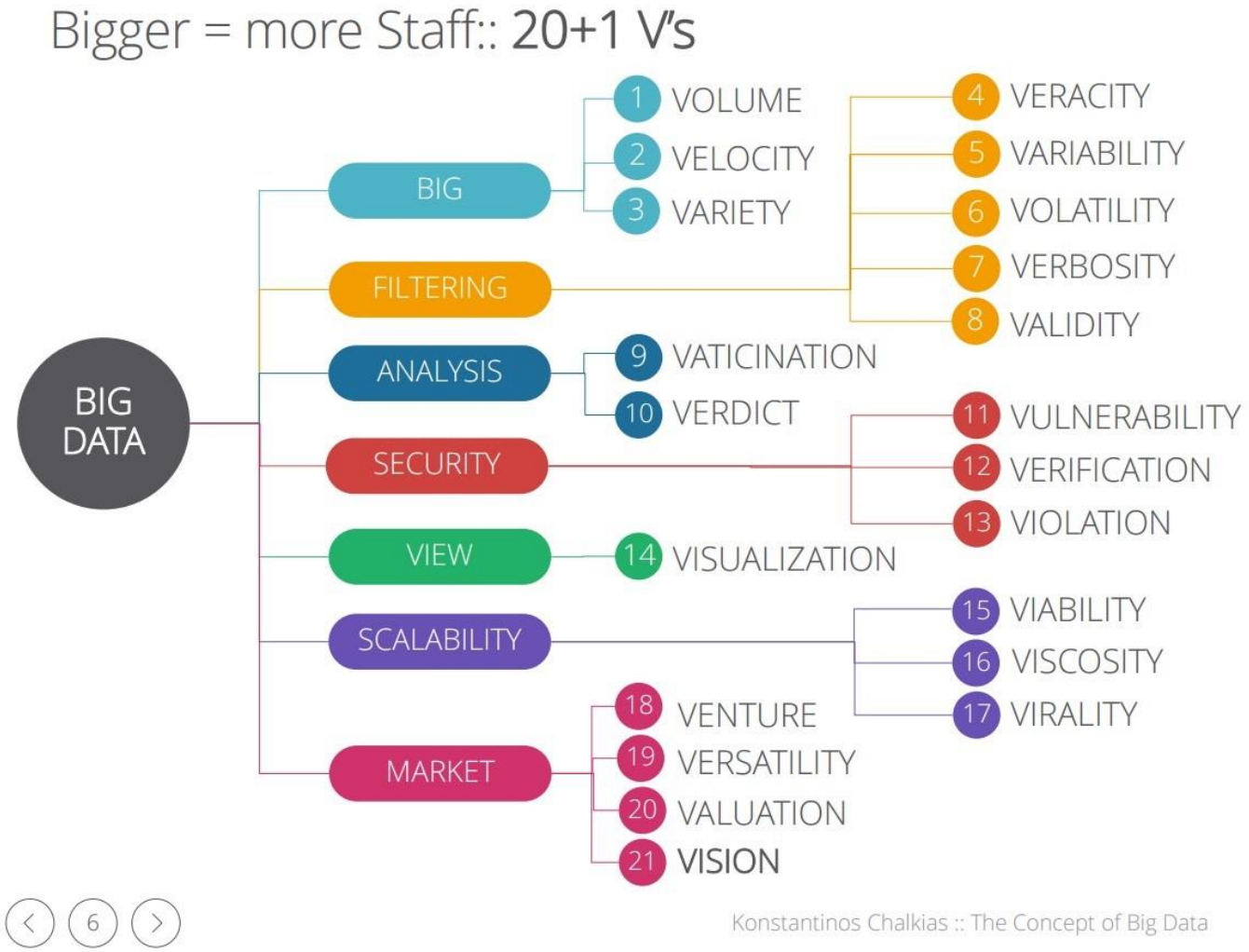
Value

- Value of data refers to the usefulness of data for the intended purpose.
- The end goal of any big data analytics system is to extract value from the data.
- The value of the data is also related to the veracity or accuracy of the data. For some applications value also depends on how fast we are able to process the data.

9 V's of Big Data



20+1 V's of Big Data



Failure of Traditional Database in Handling Big Data

- RDBMS were developed to store the data that were beyond the storage capacity of a single computer.
- The start of a new technology is always due to limitations in the older technologies and the necessity to overcome them.
- Below are the limitations of traditional database in handling big data:
  - Exponential increase in data volume, which scales in terabytes and petabytes, has turned out to become a challenge to the RDBMS in handling such a massive volume of data.
  - To address this issue, the RDBMS increased the number of processors and added more memory units, which in turn increased the cost.
  - Almost 80% of the data fetched were of semi-structured and unstructured format, which RDBMS could not deal with.
  - RDBMS could not capture the data coming in at high velocity.

Failure of Traditional Database in Handling Big Data

Differences in the attributes of Big Data and RDBMS

ATTRIBUTES	RDBMS	BIG DATA
Data volume	gigabytes to terabytes	petabytes to zettabytes
Organization	centralized	distributed
Data type	structured	unstructured and semi-structured
Hardware type	high-end model	commodity hardware
Updates	read/write many times	write once, read many times
Schema	static	dynamic

Domain Specific Examples of Big Data

- The applications of big data span a wide range of domains including (but not limited to) homes, cities, environment, energy systems, retail, logistics, industry, agriculture, Internet of Things, and healthcare.

- **Web** (Web Analytics; Performance Monitoring; Ad Targeting & Analytics; Content Recommendation (Netflix, YouTube))
- **Financial** (Credit Risk Modeling; Fraud Detection)
- **Healthcare** (Epidemiological Surveillance, …)
- **Internet of Things** (Smart Irrigation, …)
- **Environment** (Air Pollution Monitoring, Forest Fire Detection)
- **Logistics & Transportation** (Remote Vehicle Diagnostics, …)
- **Industry** (Machine Diagnosis & Prognosis, …)
- **Retail** (Customer Recommendations, Store Layout Optimization, …)

Eras of Big Data

- The evolution of Big Data and Data Analytics is often described in terms of different eras, as outlined by various authors.
- Doug Laney, a research vice president at Gartner, initially proposed the three Vs (Volume, Velocity, and Variety) as defining characteristics of Big Data.
- Subsequently, this concept has evolved to include additional Vs and distinct eras in the timeline of Big Data and Data Analytics.

1. First Era: BI and Data Warehousing (Before 2000s)

- **Characteristics**
  - Focus on structured data.
  - Relational databases and data warehouses.
  - Business Intelligence (BI) tools for reporting and querying.

- **Key Technologies**
  - Traditional databases (e.g., Oracle, SQL Server).
  - Enterprise data warehouses.
  - BI tools (e.g., Power BI, Tableau).

2. Second Era: Big Data (Mid-2000s to Early 2010s)

- **Characteristics**
  - Emergence of the three Vs: Volume, Velocity, and Variety.
  - Handling larger and more diverse datasets.
  - NoSQL databases for unstructured and semi-structured data.

- **Key Technologies**
  - Hadoop for distributed storage and processing.
  - NoSQL databases (e.g., MongoDB, Cassandra).
  - MapReduce programming model.

3. Third Era: Data Science and Advanced Analytics (Mid-2010s Onwards)

- **Characteristics**
  - Inclusion of additional Vs like Veracity and Variability.
  - Advanced analytics, machine learning, and predictive modeling.
  - Focus on extracting meaningful insights from data.

- **Key Technologies**
  - Machine learning frameworks (e.g., TensorFlow, PyTorch).
  - Advanced analytics platforms (e.g., Spark, Databricks).
  - Cloud-based analytics services (e.g., AWS Sagemaker, Azure Machine Learning).

4. Fourth Era: AI and Cognitive Analytics (Emerging)

- **Characteristics**
  - Integration of artificial intelligence (AI) and cognitive computing.
  - Emphasis on natural language processing and automated decision-making.
  - Increasing automation in analytics processes.

- **Key Technologies**
  - Natural language processing (NLP) tools.
  - AI-driven analytics platforms.
  - Automated machine learning (AutoML) tools.

Transversal themes

- **Cloud Computing**
  - The widespread adoption of cloud platforms has significantly influenced the evolution of Big Data and analytics, providing scalable and cost-effective resources.
- **Edge Computing**
  - The rise of edge computing allows data processing to occur closer to the source of data generation, addressing latency and bandwidth concerns.
- **IoT Integration**
  - The Internet of Things (IoT) has contributed to the growth of data volume and variety, with sensors and devices generating massive amounts of data.

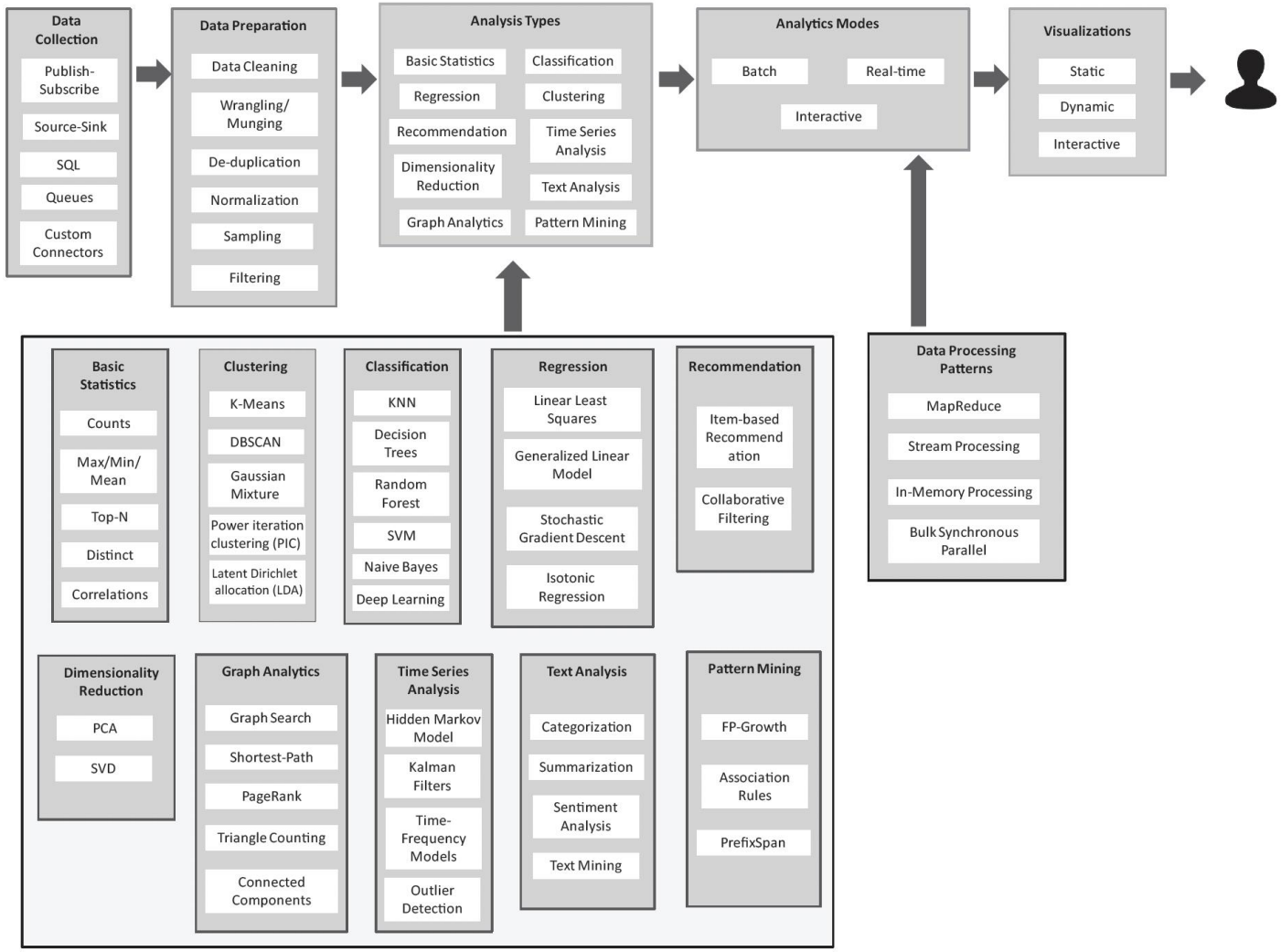
Eras Summary

- This evolution highlights the increasing complexity and capabilities of data processing and analytics.
- Each era builds upon the advancements of the previous one, incorporating new technologies and methodologies to address the challenges posed by the growing volume and diversity of data.
- The future is likely to involve further integration of AI, automation, and enhanced collaboration between humans and intelligent systems in the realm of data analytics.

Analytics Flow for Big Data

- A generic flow for big data analytics, detailing the steps involved in the implementation of a typical analytics application and the options available at each step, is presented.
- Next figure shows the analytics flow with various steps. For an application, selecting the options for each step in the analytics flow can help in determining the right tools and frameworks to perform the analyses.
  - **Data Collection**
  - **Data Preparation**
  - **Analysis Types**
  - **Analysis Modes**
  - **Visualizations**

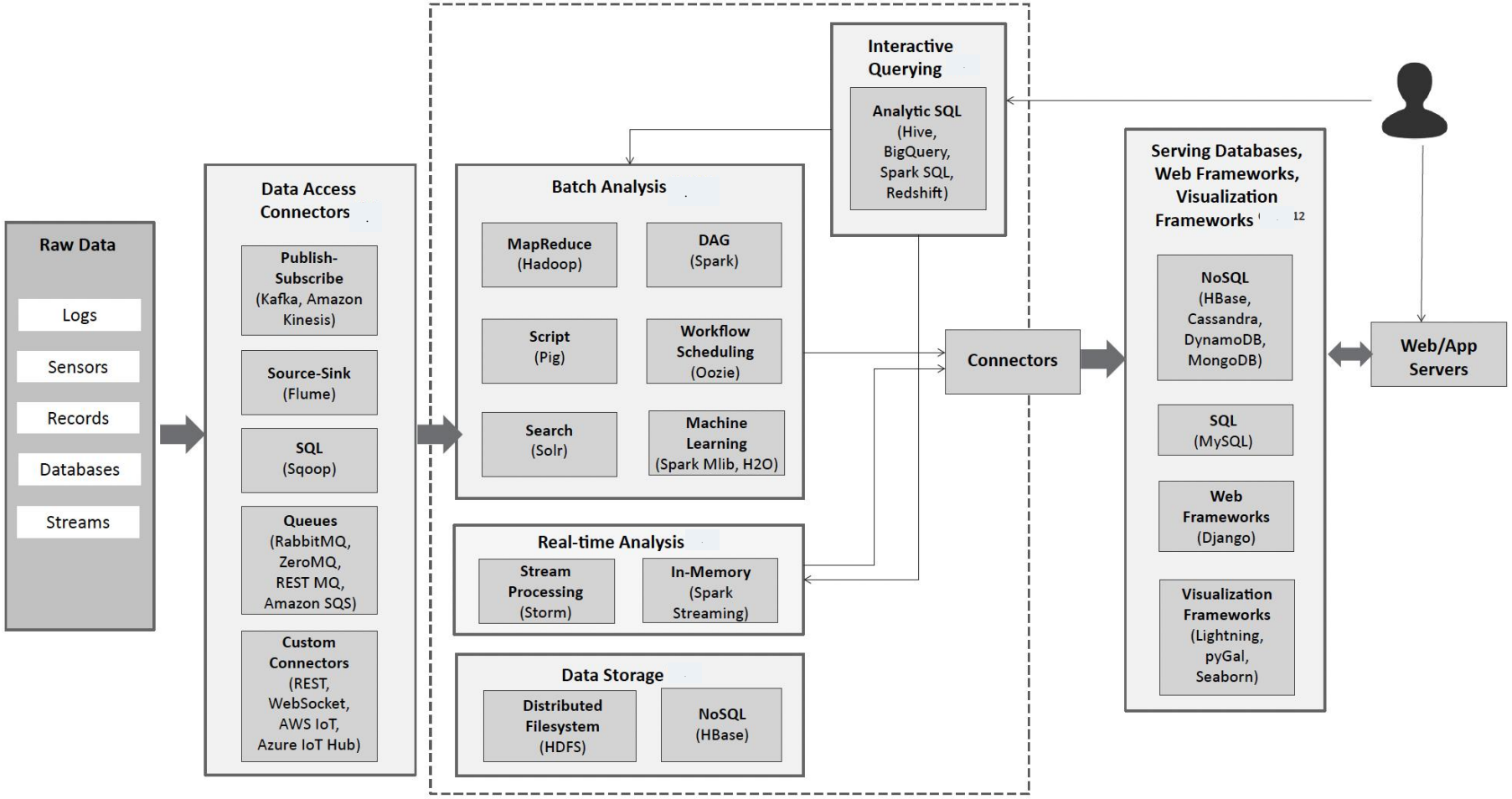
Analytics Flow for Big Data



Big Data Stack

- While the Hadoop framework has been one of the most popular frameworks for big data analytics, there are several types of computational tasks for which Hadoop does not work well.
- Next figure identify the cases where Hadoop works and where it does not and describe the motivation for having a Big Data stack that can be used for various types of analytics and computational tasks.

Big Data Stack



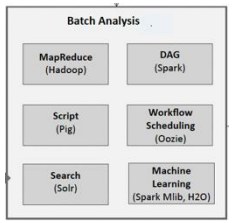
Big Data Stack

- **Hadoop** is an open source framework for distributed batch processing of massive scale data using the MapReduce programming model.
- The **MapReduce** programming model is useful for applications in which the data involved is so massive that it would not fit on a single machine.
- In such applications, the data is typically stored on a distributed file system (such as Hadoop Distributed File System - **HDFS**).
- MapReduce programs take advantage of locality of data and the data processing takes place on the nodes where the data resides. In other words, the computation is moved to where the data resides, as opposed the traditional way of moving the data from where it resides to where the computation is done.

Big Data Stack

- MapReduce is best suited for descriptive analytics and the basic statistics computational tasks because the operations involved can be done in parallel (for example, computing counts, mean, max/min, distinct, top-N, filtering and joins). Many of these operations are completed with a single MapReduce job.
- For more complex tasks, multiple MapReduce jobs can be chained together. However, when the computations are iterative in nature, where a MapReduce job has to be repeatedly run, MapReduce takes a performance hit because of the overhead involved in fetching the data from HDFS in each iteration.

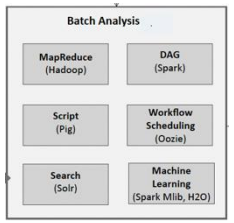
Big Data Stack: Batch Analytics (1/2)



The batch analytics block in the big data stack includes various frameworks which allow analysis of data in batches. These include the following:

- **Hadoop-MapReduce**: Hadoop is a framework for distributed batch processing of big data. The MapReduce programming model is used to develop batch analysis jobs which are executed in Hadoop clusters.
- **Pig**: Pig is a high-level data processing language which makes it easy for developers to write data analysis scripts which are translated into MapReduce programs by the Pig compiler.
- **Oozie**: Oozie is a workflow scheduler system that allows managing Hadoop jobs. With Oozie, you can create workflows which are a collection of actions (such as MapReduce jobs) arranged as Direct Acyclic Graphs (DAG).

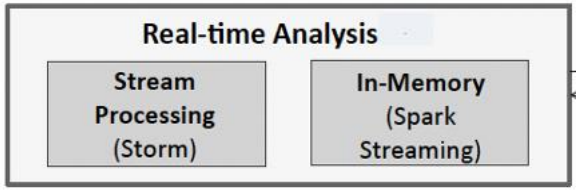
Big Data Stack: Batch Analytics (2/2)



The batch analytics block in the big data stack includes various frameworks which allow analysis of data in batches. These include the following:

- **Spark**: Apache Spark is an open source cluster computing framework for data analytics. Spark includes various high-level tools for data analysis such as Spark Streaming for streaming jobs, Spark SQL for analysis of structured data, MLlib machine learning library for Spark, and GraphX for graph processing.
- **Solr**: Apache Solr is a scalable and open-source framework for searching data.
- **Machine Learning**: Some examples are Spark MLib and H2O frameworks. Spark MLlib is the Spark's machine learning library which provides implementations of various machine learning algorithms. H2O is an open source predictive analytics framework which provides implementations of various machine learning algorithms.

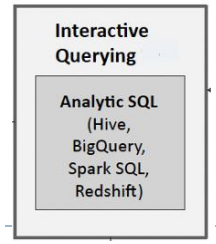
Big Data Stack: Real-time Analytics



The real-time analytics block includes the Apache Storm and Spark Streaming frameworks.

- **Apache Storm** is a framework for distributed and fault-tolerant real-time computation. Storm can be used for real-time processing of streams of data. Storm can consume data from a variety of sources such as publish-subscribe messaging frameworks (such as Kafka or Kinesis), messaging queues (such as RabbitMQ or ZeroMQ) and other custom connectors.
- **Spark Streaming** is a component of Spark which allows analysis of streaming data such as sensor data, click stream data, web server logs, for instance. The streaming data is ingested and analyzed in micro-batches. Spark Streaming enables scalable, high throughput and fault-tolerant stream processing.

Big Data Stack: Interactive Querying



- Interactive querying systems allow users to query data by writing statements in SQL-like languages. Some examples of interactive querying systems:
  - **Spark SQL**: Spark SQL is a component of Spark which enables interactive querying. Spark SQL is useful for querying structured and semi-structured data using SQL-like queries.
  - **Hive**: Apache Hive is a data warehousing framework built on top of Hadoop. Hive provides an SQL-like query language called Hive Query Language, for querying data residing in HDFS.
  - **Amazon Redshift**: Amazon Redshift is a fast, massive-scale managed data warehouse service. Redshift specializes in handling queries on datasets of sizes up to a petabyte or more parallelizing the SQL queries across all resources in the Redshift cluster.
  - **Google BigQuery**: Google BigQuery is a service for querying massive datasets. BigQuery allows querying datasets using SQL-like queries.

Big Data Stack: Serving Databases, Web & Visualization Frameworks

- While the various analytics blocks process and analyze the data, the results are stored in serving databases for subsequent tasks of presentation and visualization.
- These serving databases allow the analyzed data to be queried and presented in the web applications.
- The following SQL and NoSQL databases can be used as serving databases:
  - **MySQL**: MySQL is one of the most widely used Relational Database Management System (RDBMS) and is a good choice to be used as a serving database for data analytics applications where the data is structured.

Big Data Stack: Serving Databases, Web & Visualization Frameworks

- The following SQL and NoSQL databases can be used as serving databases (cont.):
  - **Cassandra**: Cassandra is a scalable, highly available, fault tolerant open source non-relational database system.
  - **MongoDB**: MongoDB is a document oriented non-relational database system. MongoDB is powerful, flexible and highly scalable database designed for web applications and is a good choice for a serving database for data analytics applications.
- Some examples of visualization tools and frameworks:
  - **Lightning**: Lightning is a framework for creating web-based interactive visualizations.
  - **Pygal**: The Python Pygal library is an easy to use charting library which supports charts of various types.
  - **Seaborn**: Seaborn is a Python visualization library for plotting attractive statistical plots.

Mapping Analytics Flow to Big Data Stack – Part I

Data Collection

Analysis Type	Framework (Mode)
Publish-Subscribe	Kafka, Kinesis
Source-Sink	Flume
SQL	Sqoop
Queues	SQS, RabbitMQ, ZeroMQ, RESTMQ
Custom Connectors	REST, WebSocket, MQTT

Data Preparation

Analysis Type	Framework
Data Cleaning	Open Refine
Data Wrangling	Open Refine DataWrangler
De-Duplication	Open Refine, Pig, Hive, Spark SQL
Normalization Sampling, Filtering	MapReduce, Pig, Hive, Spark SQL

Basic Statistics

Analysis Type	Framework (Mode)
Counts, Max, Min, Mean, Top-N, Distinct	Hadoop-MapReduce (Batch), Pig (Batch), Spark (Batch), Spark Streaming (Realtime), Spark SQL (Interactive), Hive (Integrative), Storm (Real-time)
Correlations	Hadoop-MapReduce (Batch), Spark Mlib (Batch)

Clustering

Analysis Type	Framework (Mode)
K-Means	Hadoop-MapReduce (Batch), Spark Mlib (Batch & Real-time) H2O (Batch)
DBSCAN	Spark (Batch)
Gaussian Mixture	Spark Mlib (Batch)
PIC	Spark Mlib (Batch)
LDA	Spark Mlib (Batch)

Classification

Analysis Type	Framework (Mode)
KNN	Spark Mlib (Batch , Realtime)
Decision Trees	Spark Mlib (Batch, Realtime)
Random Forest	Spark Mlib (Batch , Realtime), H2O (Batch)
SVM	Spark Mlib (Batch , Realtime)
Naïve Bayes	Spark Mlib (Batch, Realtime), H2O (Batch)
Deep Learning	H2O (Batch)

Regression

Analysis Type	Framework (Mode)
Linear Least Squares	Spark Mlib (Batch, Realtime)
Generalized Linear Model	H2O (Batch)
Stochastic Gradient Descent	Spark Mlib (Batch, Realtime)
Isotonic Regression	Spark Mlib (Batch, Realtime)

Mapping Analytics Flow to Big Data Stack – Part II

Graph Analytics

Analysis Type	Framework (Mode)
Graph Search	Spark GraphX (Batch)
Shortest-Path	Spark GraphX (Batch)
PageRank	Spark GraphX (Batch)
Triangle Counting	Spark GraphX (Batch)
Connected Components	Spark GraphX (Batch)

Time Series Analysis

Analysis Type	Framework (Mode)
Kalman Filter	Spark (Realtime)
Time Frequency Models	Spark (Realtime)
Outlier Detection	Storm (Realtime), Spark (Batch, Realtime)

Dimensionality Reduction

Analysis Type	Framework (Mode)
SVD	Spark Mlib (Batch)
PCA	Spark Mlib (Batch), H2O (Batch)

Recommendation

Analysis Type	Framework (Mode)
Item-bases Recommendation	Spark Mlib (Batch)
Collaborative Filtering	Spark Mlib (Batch)

Text Analysis

Analysis Type	Framework (Mode)
Categorization	Hadoop-MapReduce (Batch), Storm (Realtime), Spark (Batch, Realtime)
Summarization	Spark (Batch)
Sentiment Analysis	Storm (Realtime), Spark (Batch, Realtime)
Text Mining	Storm (Realtime), Spark (Batch, Realtime)

Pattern Mining

Analysis Type	Framework (Mode)
FP-Growth	Spark Mlib (Batch)
Association Rules	Spark Mlib (Batch)
PrefixSpan	Spark Mlib (Batch)

Visualization

Analysis Type	Framework (Mode)
Web Frameworks	Django, Flask
SQL Databases	MySQL
NoSQL Databases	Hbase, DynamoDB, Cassandra, MongoDB
Visualization Frameworks	Lightning, pyGal, Seaborn

Data analytics tools to handle the big data revolution

- Numerous custom big data analytics software are in existence and moreover, these can be constructed easily.
- The most widely used tools that help the developers to make successful big data analytics software are Hadoop, Cassandra, Plotly, Bokeh, Neo4j, and so on.











Data analytics tools to handle the big data revolution

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1. Hadoop



- Hadoop is a product from Apache.
- This programming library can handle a large number of informational indexes in multiple PCs while utilizing successful programming models.
- Hadoop is mainly preferred because of its ability to store and process huge amounts of data at a high rate.
- Hadoop's distributed computing model makes big data processing fast.
- The computing power is directly proportional to the number of nodes.
- Also as it is an open-source framework, it is free and usually uses commodity hardware to store high voluminous data.

Data analytics tools to handle the big data revolution



2. Cassandra:

- Cassandra is a NoSQL database for high speed, online transactional data.
- It provides high accessibility and versatility without affecting the execution of ware equipment and cloud foundation.
- The best part of this is that it works fluently without any chances of failure.
- It also helps in keeping the updated status of the neighboring nodes in the same cluster using the Gossip Protocol.
- Netflix and eBay use Cassandra to handle big data.

Data analytics tools to handle the big data revolution

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3. Plotly:



- Plotly provides an ecosystem of data analysis tools designed to transform Big Data exploration into interactive and scalable applications, overcoming the limitations of static dashboards and spreadsheets.
- Plotly is an interactive online visualization tool used for data analytics, scientific graphs, and other types of visualization. It consists of an API for Python too.
- It is used to draw unique representation even if when the organization does not have enough time or abilities for meeting enormous information needs.
- It utilizes online instruments to construct dazzling and useful illustrations without any extra effort. The positive thing about it is that it is easily integrated with Pandas.

Data analytics tools to handle the big data revolution



4. Bokeh:

- Bokeh is an interactive visualisation library in Python that stands out in the Big Data landscape for its ability to render large volumes of data directly in the browser with high performance.
- Bokeh is just like Plotly and used for making instructive and straightforward representations.
- It is used by information examination specialists, so that intelligent information applications can be constructed effectively and easily.
- Many specialists prefer Bokeh because it's the most progressive visual information portrayal instrument.

Data analytics tools to handle the big data revolution

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5. Neo4j:



- Neo4j is a specialized database that is widely used in managing graphs.
- Traditional database software stores information in tabular form (just like the representation in Microsoft Excel sheets) except that Neo4j is used on a much larger scale.
- It is used for storing high voluminous data but is primarily designed to capture the relationship between items of information.

Summary

- We described what is analytics and the various types of analytics applications. **Analytics** is the process of extracting and creating information from raw data by filtering, processing, categorizing, condensing and contextualizing the data. Descriptive analytics deals with analyzing past data to present it in a summarized form which can be easily interpreted. Diagnostic analytics deals with the analysis of past data to diagnose the reason for happening of certain events. Predictive analytics involves predicting the occurrence of an event or the likely outcome of an event or forecasting the future values using prediction models. Prescriptive analytics uses multiple prediction models to predict various outcomes and the best course of action for each outcome.

Summary

- **Big data analytics** deals with collection, storage, processing and analysis of massive-scale data. Big data analytics involves several steps starting from data cleansing, data munging (or wrangling), data processing and visualization. We described the characteristics of big data including **volume, velocity, variety, veracity, and value**. Domain specific examples and applications of big data were described. Next, we proposed a **generic flow for big data analytics**, detailing the steps involved in a typical analytics task and the options available at each step. Data preparation step involves various tasks such as data cleansing, data wrangling or munging, de-duplication, normalization, sampling and filtering. We described the various analysis types and the computational 'giants'. We proposed an described a **big data stack** comprising of proven and open-source big data frameworks. Finally, an **approach for mapping the analytics flow to specific tools and frameworks in the big data stack** was described.


ANY
QUESTIONS!

Present

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O modelo de IA avançada da Anthropic para coding.

Llama 4 Maverick

O modelo IA avançada criado pela Meta, eficiente e versátil.